

Determinantal point processes

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CHAPTER 1

Non-intersecting path ensembles

1. Non-intersecting paths

Let $G = (V, E)$ be a directed graph. Throughout we assume that G is acyclic, so that no directed path can visit the same vertex more than once. Each edge $e \in E$ is assigned a strictly positive weight $w(e) \in (0, \infty)$.

A (directed) path π in G is a finite sequence of vertices

$$\pi = (v_0, v_1, \dots, v_k),$$

such that $(v_{i-1}, v_i) \in E$ for all $i = 1, \dots, k$. The weight of a path is defined multiplicatively by

$$w(\pi) = \prod_{i=1}^k w(v_{i-1}, v_i).$$

Fix an integer $n \geq 1$ and choose two ordered n -tuples of distinct vertices

$$P_1, \dots, P_n \quad \text{and} \quad Q_1, \dots, Q_n$$

in V , interpreted as starting points and ending points. For $P, Q \in V$, we denote by $\Pi(P, Q)$ the set of all directed paths in G starting at P and ending at Q .

A path ensemble is an n -tuple $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$ with $\pi_j \in \Pi(P_j, Q_j)$. The ensemble is called *non-intersecting* if no two paths share a vertex, that is,

$$\pi_i \cap \pi_j = \emptyset \quad \text{for all } i \neq j,$$

where intersections are understood at the level of vertices. We write

$$\Pi_{\text{ni}}((P_1, \dots, P_n), (Q_1, \dots, Q_n))$$

for the set of all non-intersecting path ensembles connecting the prescribed starting points to the prescribed ending points.

The weight of an ensemble $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$ is defined by

$$w(\boldsymbol{\pi}) = \prod_{j=1}^n w(\pi_j).$$

Assuming that the total weight of all non-intersecting ensembles is finite, we introduce the partition function

$$Z = \sum_{\boldsymbol{\pi} \in \Pi_{\text{ni}}} w(\boldsymbol{\pi}),$$

and define a probability measure on Π_{ni} by

$$\mathbb{P}(\boldsymbol{\pi}) = \frac{1}{Z} w(\boldsymbol{\pi}).$$

A key role in the analysis of non-intersecting paths is played by the single-path transition function g defined by

$$g(P, Q) = \sum_{\pi \in \Pi(P, Q)} w(\pi), \quad P, Q \in V,$$

which record the total weight of all directed paths between two vertices. The fundamental connection between single-path and multi-path quantities is provided by the Lindström–Gessel–Viennot theorem.

THEOREM 1 (Lindström–Gessel–Viennot). *Let $G = (V, E)$ be a directed acyclic graph with edge-weights $w: E \rightarrow (0, \infty)$. Fix vertices $P_1, \dots, P_n$ and $Q_1, \dots, Q_n$ in V . Then*

$$(1) \quad \det[g(P_i, Q_j)]_{i,j=1}^n = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \sum_{\pi \in \Pi_{\text{ni}}((P_1, \dots, P_n), (Q_{\sigma(1)}, \dots, Q_{\sigma(n)}))} w(\pi).$$

PROOF. Expanding the determinant gives

$$\det[g(P_i, Q_j)]_{i,j=1}^n = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n g(P_i, Q_{\sigma(i)}).$$

Using the definition of g and distributivity, we obtain

$$\prod_{i=1}^n g(P_i, Q_{\sigma(i)}) = \sum_{(\pi_1, \dots, \pi_n)} \prod_{i=1}^n w(\pi_i),$$

where the sum runs over all n -tuples with $\pi_i \in \Pi(P_i, Q_{\sigma(i)})$. Thus the determinant equals a signed sum over all such path families. Note that this include both families of paths that are non-intersecting and intersecting, and it remains to show that the sum over intersecting paths vanishes. To this end we construct a sign-reversing involution on the set of intersecting families.

Since G is acyclic, fix a topological ordering of its vertices. Given an intersecting family, let v be the earliest vertex (in this order) that lies on at least two paths, and let $a < b$ be the smallest indices such that $v \in \pi_a \cap \pi_b$. Write

$$\pi_a = \alpha_a \cdot \beta_a, \quad \pi_b = \alpha_b \cdot \beta_b,$$

where α_a, α_b are the initial segments ending at v and β_a, β_b are the remaining tails. Define new paths by swapping the tails,

$$\pi'_a = \alpha_a \cdot \beta_b, \quad \pi'_b = \alpha_b \cdot \beta_a,$$

and let $\sigma' = \sigma \circ (ab)$. This preserves total weight and reverses the sign, while remaining an involution. Hence all intersecting families cancel pairwise, and only non-intersecting ensembles contribute, yielding (1). $\square$

COROLLARY 2 (Partition function). *Assume that any non-intersecting path ensemble connecting $\{P_1, \dots, P_n\}$ to $\{Q_1, \dots, Q_n\}$ necessarily matches P_i to Q_i for each i . Then*

$$Z = \det[g(P_i, Q_j)]_{i,j=1}^n.$$

PROOF. Under the stated assumption, all terms in (1) corresponding to non-trivial permutations vanish, and the remaining term is precisely the partition function Z . $\square$

2. Graphs obtained by gluing columns

Fix an integer $N \geq 1$. We consider directed graphs whose vertex set is the full integer strip

$$V = \{0, 1, \dots, N\} \times \mathbb{Z}.$$

Edges are added column-by-column. More precisely, for each $x \in \{0, 1, \dots, N-1\}$ we specify a set of directed edges between the slices $\{x\} \times \mathbb{Z}$ and $\{x+1\} \times \mathbb{Z}$, and possibly also vertical edges inside the right slice $\{x+1\} \times \mathbb{Z}$. Gluing the columns means that the slice $\{x+1\} \times \mathbb{Z}$ serves simultaneously as the right boundary of column x and the left boundary of column $x+1$.

We will use four elementary column types. In all cases $y \in \mathbb{Z}$.

Type I (Bernoulli up-step). The only edges in column x are

$$(x, y) \rightarrow (x+1, y), \quad (x, y) \rightarrow (x+1, y+1).$$

Thus a path crossing the column either goes straight or steps up by one.

Type II (Bernoulli down-step). The only edges in column x are

$$(x, y) \rightarrow (x+1, y), \quad (x, y) \rightarrow (x+1, y-1).$$

Thus a path crossing the column either goes straight or steps down by one.

Type III (horizontal + vertical down at the right slice). The column contains all horizontal edges

$$(x, y) \rightarrow (x+1, y),$$

and, in addition, all vertical edges at the right slice pointing down,

$$(x+1, y) \rightarrow (x+1, y-1).$$

Equivalently, after entering the slice $x+1$, a path may move down by an arbitrary number of steps before proceeding further to the right in subsequent columns.

Type IV (horizontal + vertical up at the right slice). The column contains all horizontal edges

$$(x, y) \rightarrow (x+1, y),$$

and, in addition, all vertical edges at the right slice pointing up,

$$(x+1, y) \rightarrow (x+1, y+1).$$

Equivalently, after entering the slice $x+1$, a path may move up by an arbitrary number of steps before proceeding further to the right.

By choosing for each $x \in \{0, \dots, N-1\}$ one of the four types above, we obtain a directed graph on $\{0, \dots, N\} \times \mathbb{Z}$ built by gluing columns. Since every edge either increases x by one (horizontal/diagonal) or keeps x fixed while changing y (vertical edges within a slice), and since vertical edges are oriented consistently (up or down within a given slice), the resulting graph is directed and acyclic.

EXAMPLE 3 (Alternating Type I and Type III columns). *As a concrete illustration of the column-gluing construction, consider a graph on*

$$V = \{0, 1, \dots, N\} \times \mathbb{Z}$$

obtained by gluing columns in an alternating pattern: for even x we use Type I on the strip from x to $x+1$, and for odd x we use Type III.

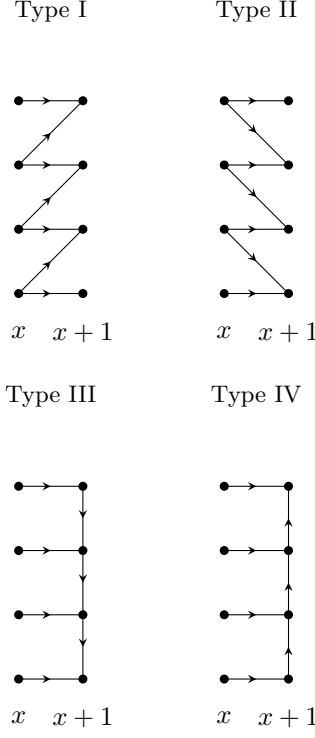


FIGURE 1. Four elementary column types used to build directed graphs on $\{0, \dots, N\} \times \mathbb{Z}$.

EXAMPLE 4 (N columns of Type IV followed by N columns of Type III). Fix an integer $N \geq 1$ and consider the directed graph on

$$V = \{0, 1, \dots, 2N\} \times \mathbb{Z}$$

obtained by gluing $2N$ columns as follows: for $x = 0, 1, \dots, N-1$ the column between x and $x+1$ is of Type IV, and for $x = N, N+1, \dots, 2N-1$ the column between x and $x+1$ is of Type III. In other words, we use a block of N Type IV columns followed by a block of N Type III columns. This produces a directed acyclic graph with a natural left-to-right direction and a single interface at $x = N$.

2.1. Edge weights. We now equip the glued graph with edge weights. Throughout, we impose the convention that *all horizontal edges have weight 1*. Only the remaining (non-horizontal) edges carry nontrivial weights, which may depend on the position (x, y) and on the column type.

Consider a column between the slices x and $x+1$, and let $y \in \mathbb{Z}$.

Type I. The diagonal up-edges

$$(x, y) \rightarrow (x+1, y+1)$$

are assigned weights

$$w((x, y) \rightarrow (x+1, y+1)) = \alpha_{x,y} \in (0, \infty),$$

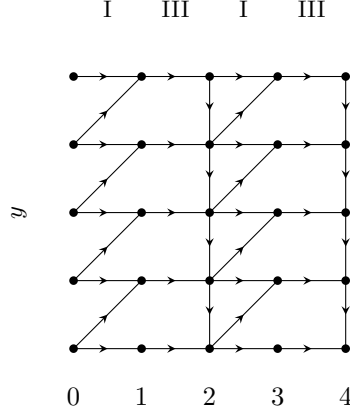


FIGURE 2. An example obtained by gluing columns in the alternating pattern I–III–I–III. Type I columns allow $(x, y) \rightarrow (x+1, y)$ and $(x, y) \rightarrow (x+1, y+1)$, while Type III columns allow $(x, y) \rightarrow (x+1, y)$ and vertical down moves $(x+1, y) \rightarrow (x+1, y-1)$ within the right slice.

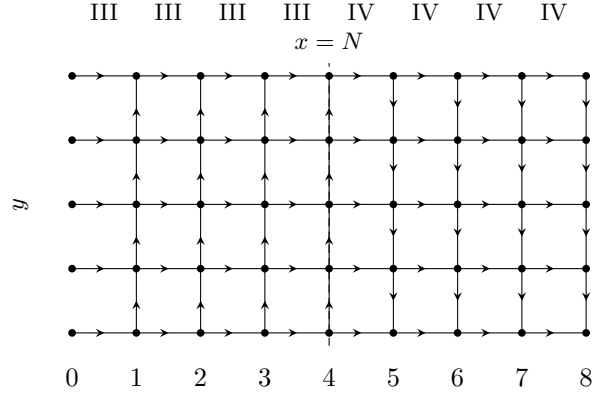


FIGURE 3. Example with $N = 4$: a block of four Type III columns followed by four Type IV columns.

while the horizontal edges $(x, y) \rightarrow (x+1, y)$ have weight 1.

Type II. The diagonal down-edges

$$(x, y) \rightarrow (x+1, y-1)$$

are assigned weights

$$w((x, y) \rightarrow (x+1, y-1)) = \beta_{x,y} \in (0, \infty),$$

while the horizontal edges $(x, y) \rightarrow (x+1, y)$ have weight 1.

Type III. The vertical up-edges at the right slice,

$$(x+1, y) \rightarrow (x+1, y+1),$$

are assigned weights

$$w((x+1, y) \rightarrow (x+1, y+1)) = \gamma_{x+1, y} \in (0, \infty),$$

while the horizontal edges $(x, y) \rightarrow (x+1, y)$ have weight 1.

Type IV. The vertical down-edges at the right slice,

$$(x+1, y) \rightarrow (x+1, y-1),$$

are assigned weights

$$w((x+1, y) \rightarrow (x+1, y-1)) = \delta_{x+1, y} \in (0, \infty),$$

while the horizontal edges $(x, y) \rightarrow (x+1, y)$ have weight 1.

With these conventions, the weight of a path is the product of the weights of its edges, and the weight of a path ensemble is the product of the weights of the individual paths. Since all horizontal edges have unit weight, only diagonal and vertical moves contribute nontrivially to the total weight.

2.2. Non-interesting path ensembles and a point process. We impose boundary conditions

$$P_j = (0, -j), \quad Q_j = (2N, -j), \quad j = 1, \dots, n,$$

and consider n -tuples of non-intersecting directed paths connecting P_j to Q_j .

The paths naturally induce a point process as follows. Let $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$ be a non-intersecting path ensemble. At each vertical section $x \in \{0, 1, \dots, 2N\}$ we associate an ordered n -tuple of heights

$$\boldsymbol{\eta}^{(x)} = (\eta_1^{(x)} < \dots < \eta_n^{(x)}) \in \mathbb{Z}^n,$$

defined according to the type of column immediately to the left of x .

- If the column between $x-1$ and x is of **Type III**, then $\eta_j^{(x)}$ is the *highest* height reached by the j th path at slice x .
- If the column between $x-1$ and x is of **Type IV**, then $\eta_j^{(x)}$ is the *lowest* height reached by the j th path at slice x .
- If the column between $x-1$ and x is of **Type I or II**, then no vertical motion occurs at slice x , and $\eta_j^{(x)}$ is the unique height at which the j th path visits that slice.

By non-intersection, the ordering

$$\eta_1^{(x)} < \dots < \eta_n^{(x)}$$

is preserved for all x . The boundary conditions read

$$\boldsymbol{\eta}^{(0)} = (-n, \dots, -1), \quad \boldsymbol{\eta}^{(2N)} = (-n, \dots, -1).$$

The associated point process. The non-intersecting path ensemble induces a space-time point configuration

$$\eta := \{(x, \eta_j^{(x)}) : x = 0, \dots, 2N, j = 1, \dots, n\} \subset \{0, \dots, 2N\} \times \mathbb{Z}.$$

We refer to η as the point process associated with the non-intersecting paths. This is the main observable of interest.

Transition matrices. For each $x \in \{0, 1, \dots, 2N - 1\}$ we define a (generally infinite) transition matrix

$$T_{x,x+1} = (T_{x,x+1}(y, y'))_{y, y' \in \mathbb{Z}},$$

where $T_{x,x+1}(y, y')$ is the total weight of all single-path segments whose *recorded height* changes from y at slice x to y' at slice $x + 1$ while crossing the x th column.

For $0 \leq x < x' \leq 2N$ we define the multi-step transition matrix by

$$T_{x,x'} := T_{x,x+1} T_{x+1,x+2} \cdots T_{x'-1,x'}.$$

Determinantal distribution of the point process.

THEOREM 5 (Product form of the path measure). *For any choice of strictly increasing n -tuples*

$$\boldsymbol{\eta}^{(1)}, \dots, \boldsymbol{\eta}^{(2N-1)} \in \mathbb{Z}^n,$$

the probability that the associated point process η satisfies $\boldsymbol{\eta}^{(x)} = (\eta_1^{(x)}, \dots, \eta_n^{(x)})$ for all $x = 0, \dots, 2N$ is given by

$$\mathbb{P}(\boldsymbol{\eta}^{(1)}, \dots, \boldsymbol{\eta}^{(2N-1)}) = \frac{1}{Z} \prod_{x=0}^{2N-1} \det[T_{x,x+1}(\eta_i^{(x)}, \eta_j^{(x+1)})]_{i,j=1}^n,$$

where the partition function is

$$Z = \det[T_{0,2N}(-i, -j)]_{i,j=1}^n.$$

COROLLARY 6 (Single-slice marginal of the point process). *Fix $m \in \{0, 1, \dots, 2N\}$. For any strictly increasing n -tuple $\mathbf{y} = (y_1 < \dots < y_n)$,*

$$\mathbb{P}(\boldsymbol{\eta}^{(m)} = \mathbf{y}) = \frac{1}{Z} \det[T_{0,m}(-i, y_j)]_{i,j=1}^n \det[T_{m,2N}(y_i, -j)]_{i,j=1}^n.$$

2.3. Correlation functions. We now turn to correlation functions of the point process

$$\eta = \{(x, \eta_j^{(x)}) : x = 0, \dots, 2N, j = 1, \dots, n\} \subset \{0, \dots, 2N\} \times \mathbb{Z}.$$

For a fixed slice $m \in \{0, \dots, 2N\}$ we write

$$\eta^{(m)} := \{\eta_1^{(m)}, \dots, \eta_n^{(m)}\} \subset \mathbb{Z},$$

so that $\eta = \bigcup_{m=0}^{2N} \{m\} \times \eta^{(m)}$.

Throughout we use the transition matrices $T_{x,x'}$ defined earlier, and we set

$$G := (G_{i,j})_{i,j=1}^n, \quad G_{i,j} := T_{0,2N}(-i, -j).$$

By the LGV theorem and our boundary conditions, the partition function equals

$$Z = \det G.$$

In particular, we assume $\det G \neq 0$, so that G^{-1} exists.

Correlation functions. For $k \geq 1$, the k -point correlation function of the slice process $\eta^{(m)}$ is defined by

$$\rho_k^{(m)}(y_1, \dots, y_k) := \mathbb{P}(\{y_1, \dots, y_k\} \subset \eta^{(m)}), \quad y_1, \dots, y_k \in \mathbb{Z} \text{ distinct.}$$

Similarly, for the full space-time process η we define

$$\rho_k((x_1, y_1), \dots, (x_k, y_k)) := \mathbb{P}(\{(x_1, y_1), \dots, (x_k, y_k)\} \subset \eta),$$

for distinct space-time points $(x_a, y_a) \in \{0, \dots, 2N\} \times \mathbb{Z}$.

THEOREM 7 (Determinantal correlations on a fixed slice). *Fix $m \in \{0, 1, \dots, 2N\}$ and define a kernel on $\mathbb{Z} \times \mathbb{Z}$ by*

$$(2) \quad K_n^{(m)}(y, y') := \sum_{i,j=1}^n T_{m,2N}(y, -j) (G^{-1})_{j,i} T_{0,m}(-i, y').$$

Then the point process $\eta^{(m)}$ on $\mathbb{Z}$ is determinantal with kernel $K^{(m)}$, i.e. for any $k \geq 1$ and any distinct $y_1, \dots, y_k \in \mathbb{Z}$,

$$\rho_k^{(m)}(y_1, \dots, y_k) = \det[K_N^{(m)}(y_a, y_b)]_{a,b=1}^k.$$

PROOF. The weight-sum formula for the single-slice marginal (proved earlier) reads

$$\mathbb{P}(\eta^{(m)} = \{y_1 < \dots < y_n\}) = \frac{1}{\det G} \det[T_{0,m}(-i, y_j)]_{i,j=1}^n \det[T_{m,2N}(y_i, -j)]_{i,j=1}^n.$$

This is an instance of the Eynard–Mehta (or Cauchy–Binet) structure: a probability measure on n -point subsets of $\mathbb{Z}$ proportional to a product of two determinants. A standard determinantal calculation (expanding minors and using Cauchy–Binet with the normalization $\det A$) shows that all inclusion probabilities are given by determinants with kernel (2). $\square$

The previous theorem describes the correlations within a single slice. We next state the determinantal structure of the full space-time process η .

THEOREM 8 (Extended determinantal point process). *Define a kernel on $(\{0, \dots, 2N\} \times \mathbb{Z})^2$ by*

$$(3) \quad K_n((x, y), (x', y')) := -\mathbf{1}_{\{x < x'\}} T_{x,x'}(y, y') + \sum_{i,j=1}^n T_{x,2N}(y, -j) (G^{-1})_{j,i} T_{0,x'}(-i, y').$$

Then η is a determinantal point process with correlation kernel K , i.e. for any $k \geq 1$ and any distinct space-time points $(x_1, y_1), \dots, (x_k, y_k)$,

$$\rho_k((x_1, y_1), \dots, (x_k, y_k)) = \det[K_n((x_a, y_a), (x_b, y_b))]_{a,b=1}^k.$$

Moreover, the single-slice kernel of Theorem 7 is obtained by restriction: for $x = x' = m$ one has

$$K_n((m, y), (m, y')) = K_n^{(m)}(y, y').$$

PROOF. This is the standard Eynard–Mehta theorem for non-intersecting path ensembles with fixed boundary data, applied to the product-of-determinants representation of the path measure in terms of the one-step matrices $T_{x,x+1}$. The indicator term $-\mathbf{1}_{\{x < x'\}} T_{x,x'}$ encodes the directed time-ordering of the paths, while

the second term imposes the boundary conditions through the matrix G^{-1} . The restriction to a single slice follows immediately from (3) because the indicator term vanishes when $x = x'$. $\square$

3. Column-homogeneous parameters and Laurent transition matrices

We now specialize to the case in which all parameters are constant within each column. More precisely, for each column index r we assume that the non-horizontal edge weights do not depend on the vertical coordinate y , but only on r and on the column type. Thus we write

$$\alpha_{r,y} \equiv \alpha_r, \quad \beta_{r,y} \equiv \beta_r, \quad \gamma_{r,y} \equiv \gamma_r, \quad \delta_{r,y} \equiv \delta_r,$$

whenever the corresponding edge type is present in column r . Horizontal edges continue to have weight 1.

Under this assumption, all transition matrices $T_{r,r+1}$ are translation invariant in the vertical direction. Consequently, each $T_{r,r+1}$ is a Laurent matrix, i.e. a bi-infinite matrix whose entries depend only on the difference $y' - y$.

One-step transition matrices. Writing $k = y' - y$, the matrix elements take the following form.

- **Type I:**

$$T_{r,r+1}(y, y') = \delta_{k,0} + \alpha_r \delta_{k,1}.$$

- **Type II:**

$$T_{r,r+1}(y, y') = \delta_{k,0} + \beta_r \delta_{k,-1}.$$

- **Type III:**

$$T_{r,r+1}(y, y') = \begin{cases} \gamma_r^{y'-y}, & k \geq 0, \\ 0, & k < 0. \end{cases}$$

- **Type IV:**

$$T_{r,r+1}(y, y') = \begin{cases} \delta_r^{y-y'}, & k \leq 0, \\ 0, & k > 0. \end{cases}$$

In each case $T_{r,r+1}(y, y')$ depends only on $k = y' - y$, confirming that $T_{r,r+1}$ is a Laurent matrix.

Symbol representation. We then introduce the symbols:

$$a_r(z) := \sum_{k \in \mathbb{Z}} T_{r,r+1}(0, k) z^k, \quad z \in \mathbb{C}^\times.$$

For the four column types this gives

- **Type I:**

$$a_r(z) = 1 + \alpha_r z.$$

- **Type II:**

$$a_r(z) = 1 + \beta_r / z.$$

- **Type III:**

$$a_r(z) = \sum_{k \geq 0} \delta_r^k z^k = \frac{1}{1 - \gamma_r z},.$$

- **Type IV:**

$$a_r(z) = \sum_{k \geq 0} \gamma_r^k z^{-k} = \frac{1}{1 - \delta_r / z}.$$

Multi-step transitions. Since products of Laurent matrices are again Laurent matrices, all multi-step transition matrices

$$T_{x,x'} = T_{x,x+1} T_{x+1,x+2} \cdots T_{x'-1,x'}$$

are Laurent matrices as well. Their symbols are given by

$$a_{x,x'}(z) = \prod_{r=x}^{x'-1} a_r(z),$$

and thus

$$T_{x,x'} = L(\tilde{a}_{x,x'}),$$

where $L(a)$ stands for the Laurent matrices with symbol a , and we also used the notation $\tilde{a}(z) = a(1/z)$ -

Note also, for $i, j = 1, \dots, n$ we have

$$G_{i,j} = (L(\tilde{a}_{x,x'}))_{-i,-j} = (L(a_{x,x'}))_{i,j}$$

and thus

$$G_{i,j} = T_n(a).$$

This Laurent-symbol representation will be used extensively in the analysis of correlation kernels and scaling limits.

3.1. Taking the limit $n \rightarrow \infty$. For finite n the model is still quite involved, since the correlation kernel contains the inverse of a finite Toeplitz matrix. In general, inverting such matrices is a difficult task. A major simplification occurs, however, when we take the limit $n \rightarrow \infty$. In this regime, the finite Toeplitz matrix is replaced by a Toeplitz operator, whose inversion is, remarkably, much more tractable.

What follows requires justification—which will be provided later—but for the moment we proceed formally. We take the limit $n \rightarrow \infty$ and simply replace $T_n(a)^{-1}$ by $T(a)^{-1}$. This leads to the following result.

THEOREM 9. *Suppose that $a_{0,N}$ admits a Wiener–Hopf factorization*

$$a_{0,N} = a_+ a_-.$$

Then the limit $n \rightarrow \infty$ of the point process exists, is determinantal, and has correlation kernel

$$(4) \quad K((x, y), (x', y')) = -\mathbf{1}_{\{x < x'\}} \frac{1}{2\pi i} \oint_{|z|=1} a_{x,x'}(z) \frac{dz}{z^{y'-y+1}} \\ + \frac{1}{(2\pi i)^2} \oint_{|z|=r_z} dz \oint_{|w|=r_+} dw a_{x,2N}(z) a_+^{-1}(z) a_-^{-1}(w) a_{0,x'}(w) \frac{1}{w(w-z)} \frac{z^y}{w^{y'}},$$

where $r_{z,w}$ are such that $r_z < 1 < r_w$ and both $a(z)$ and $a(z)^{-1}$ are analytic in the annulus $r_z < |z| < r_w$.

PROOF. By definition,

$$(5) \quad K((x, y), (x', y')) = -\mathbf{1}_{\{x < x'\}} T_{x,x'}(y, y') + \sum_{i,j=0}^{\infty} T_{x,2N}(y, -j) (T(a)^{-1})_{j,i} T_{0,x'}(-i, y').$$

Using the Wiener–Hopf inversion formula

$$T(a)^{-1} = T(a_+^{-1}) T(a_-^{-1}),$$

together with

$$T_{x,2N}(y, -j) = (T(a_{x,2N}))_{-y,j}, \quad T_{0,x'}(-i, y') = (T(a_{0,x'}))_{i,-y'},$$

we obtain

(6)

$$K((x, y), (x', y')) = -\mathbf{1}_{\{x < x'\}} T_{x,x'}(y, y') + (T(a_{x,2N}) T(a_+^{-1}) T(a_-^{-1}) T(a_{0,x'}))_{-y,-y'}.$$

Since a_+^{-1} (resp. a_-^{-1}) has only nonnegative (resp. nonpositive) Fourier modes, this simplifies to

$$(7) \quad K((x, y), (x', y')) = -\mathbf{1}_{\{x < x'\}} T_{x,x'}(y, y') + (T(a_{x,2N} a_+^{-1}) T(a_-^{-1} a_{0,x'}))_{-y,-y'}.$$

Writing out the Toeplitz operators gives

$$\sum_{\ell=1}^{\infty} \frac{1}{(2\pi i)^2} \oint dz \oint dw a_{x,2N}(z) a_+^{-1}(z) a_-^{-1}(w) a_{0,x'}(w) \frac{1}{z^{-y-\ell+1}} \frac{1}{w^{\ell+y'+1}}.$$

For $|z| < |w|$,

$$\sum_{\ell=1}^{\infty} \frac{z^\ell}{w^{\ell+1}} = \frac{z}{w(w-z)},$$

which yields

$$(8) \quad K((x, y), (x', y')) = -\mathbf{1}_{\{x < x'\}} T_{x,x'}(y, y') + \frac{1}{(2\pi i)^2} \oint_{|z|=r_z} dz \oint_{|w|=r_w} dw a_{x,2N}(z) a_+^{-1}(z) a_-^{-1}(w) a_{0,x'}(w) \frac{1}{w(w-z)} \frac{z^y}{w^{y'}}.$$

Finally,

$$T_{x,x'}(y, y') = \frac{1}{2\pi i} \oint_{|z|=1} a_{x,x'}(z) \frac{dz}{z^{y'-y+1}},$$

which completes the formal computation. $\square$

3.2. Example: Aztec diamond. Consider now the case

$$a_{2x} = 1 + az, \quad x = 0, \dots, N-1,$$

and

$$a_{2x+1} = \frac{1}{1-a/z}, \quad x = 0, \dots, N-1,$$

where $a \in (0, 1)$. This corresponds to alternating columns of type I and type IV.

Configurations of the associated non-intersecting paths encode domino tilings of the Aztec diamond. Thus, we obtain a probability measure on the set of all such tilings. Vertical dominoes have weight a , while horizontal dominoes have weight 1; this model is known as the biased Aztec diamond.

The resulting point process is determinantal, with correlation kernel

(9)

$$\begin{aligned} K((2p+\varepsilon, y), (2q+\varepsilon', y')) &= -\mathbf{1}_{\{p < q\}} \frac{1}{2\pi i} \oint_{|z|=1} (1+az)^{q-p+\varepsilon'-\varepsilon} \left(1-\frac{a}{z}\right)^{-(q-p)} \frac{dz}{z^{y'-y+1}} \\ &\quad + \frac{1}{(2\pi i)^2} \oint_{|z|=r_z} dz \oint_{|w|=r_w} dw (1+az)^{-p} \left(1-\frac{a}{z}\right)^{-(N-p)} \\ &\quad \times (1+aw)^q \left(1-\frac{a}{w}\right)^{N-q} \frac{1}{z(z-w)} \frac{z^y}{w^{y'}}, \end{aligned}$$

where $a < r_z < r_w < 1/a$. By taking the limit $a \rightarrow 1$, we obtain the correlation kernel for the determinantal point process corresponding to the uniform measure on all possible tilings.

(10)

$$\begin{aligned} K((2p + \varepsilon, y), (2q + \varepsilon', y')) &= -\mathbf{1}_{\{p < q\}} \frac{1}{2\pi i} \oint_{|z|=1} (1+z)^{q-p+\varepsilon'-\varepsilon} \left(1 - \frac{1}{z}\right)^{-(q-p)} \frac{dz}{z^{y'-y+1}} \\ &\quad + \frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw (1+1z)^{-p} \left(1 - \frac{1}{z}\right)^{-(N-p)} \\ &\quad \times (1+w)^q \left(1 - \frac{1}{w}\right)^{N-q} \frac{1}{z(z-w)} \frac{z^y}{w^{y'}}, \end{aligned}$$

where γ_z is any contour around $z = 0, 1$, but not $z = -1$, and γ_w is a contour that goes around z .

3.3. Edrei–Thoma theory and natural Toeplitz building blocks. In the column-homogeneous setting, each one-step transition matrix $T_{x,x+1}$ is a Toeplitz (Laurent) matrix, so single-path weights are translation invariant in the vertical direction. In order for the Lindström–Gessel–Viennot determinants to define non-negative weights for non-intersecting path ensembles, it is natural to impose a stronger structural condition on these transitions, namely *total nonnegativity*: all minors of the Toeplitz kernel $T_{x,x+1}(y, y')$ are nonnegative. In the Toeplitz case this property is equivalent to the classical Pólya frequency condition PF_∞ .

A fundamental classification theorem due to Edrei and Thoma describes all such totally nonnegative Toeplitz kernels in terms of their symbols. In the formulation relevant for our purposes, if

$$a(z) = \sum_{k \in \mathbb{Z}} a_k z^k$$

is the symbol of a Toeplitz kernel with (a_k) totally nonnegative, then a admits a canonical factorization of the form

$$a(z) = C z^\kappa \exp(t(z + z^{-1})) \prod_{m \geq 1} \frac{1 + \alpha_m z}{1 - \gamma_m z} \prod_{m \geq 1} \frac{1 + \beta_m z^{-1}}{1 - \delta_m z^{-1}},$$

where $C \geq 0$, $\kappa \in \mathbb{Z}$, $t \geq 0$, and the parameters $\alpha_m, \beta_m, \gamma_m, \delta_m \geq 0$ satisfy appropriate summability conditions ensuring convergence on an annulus containing the unit circle.

Each factor in this representation has a direct interpretation in terms of elementary graph structures. The linear factors

$$1 + \alpha z \quad \text{and} \quad 1 + \beta z^{-1}$$

correspond to Bernoulli up- and down-steps (Types I and II), while the geometric factors

$$(1 - \delta z)^{-1} \quad \text{and} \quad (1 - \gamma z^{-1})^{-1}$$

correspond to stacks of vertical moves at a fixed slice (Types IV and III). In this way, the four elementary column types introduced earlier arise as the atomic factors in the Edrei–Thoma classification.

From this perspective, the column-gluing construction is not ad hoc: in the translation-invariant setting, any homogeneous one-step transition compatible with total nonnegativity — and hence with a positive non-intersecting path measure —

can be assembled, up to harmless shifts and exponential factors, from these elementary building blocks. The exponential term $\exp(t(z + z^{-1}))$ may be interpreted as a continuous-time nearest-neighbor walk and can be approximated by suitable limits of alternating Bernoulli columns.

Consequently, measures on non-intersecting paths constructed from products of Toeplitz determinants and satisfying the natural positivity and consistency requirements are expected to arise from graphs built out of these elementary column types. The Edrei–Thoma theorem thus provides a structural justification for the graphical framework used throughout these notes.

CHAPTER 2

Saddle point methods

1. Warm-up computations

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, consider the integral

$$I_n(f) = \int_{\mathbb{R}} f(x) e^{-nx^2/2} dx.$$

For simplicity, assume that f is smooth and bounded.

When n becomes large, the Gaussian factor becomes sharply peaked around 0. We split the integral into two parts:

$$I_n(f) = \int_{\mathbb{R} \setminus [-n^{-1/2+\varepsilon}, n^{-1/2+\varepsilon}]} f(x) e^{-nx^2/2} dx + \int_{-n^{-1/2+\varepsilon}}^{n^{-1/2+\varepsilon}} f(x) e^{-nx^2/2} dx,$$

where $0 < \varepsilon < 1/2$ is fixed.

A simple estimate shows that

$$\int_{\mathbb{R} \setminus [-n^{-1/2+\varepsilon}, n^{-1/2+\varepsilon}]} f(x) e^{-nx^2/2} dx = \mathcal{O}\left(e^{-n^{2\varepsilon}}\right), \quad n \rightarrow \infty,$$

so this contribution is exponentially small.

For the main contribution, we perform the change of variables $u = \sqrt{n}x$:

$$\int_{-n^{-1/2+\varepsilon}}^{n^{-1/2+\varepsilon}} f(x) e^{-nx^2/2} dx = \frac{1}{\sqrt{n}} \int_{-n^\varepsilon}^{n^\varepsilon} f\left(\frac{u}{\sqrt{n}}\right) e^{-u^2/2} du.$$

Using the smoothness of f , we obtain

$$I_n(f) = \frac{\sqrt{2\pi}}{\sqrt{n}} f(0) (1 + \mathcal{O}(n^{-1/2})), \quad n \rightarrow \infty.$$

This provides a simple explanation of the fact that

$$\sqrt{\frac{n}{2\pi}} e^{-nx^2/2}$$

converges, in the sense of distributions, to the Dirac delta function at 0.

We now replace the quadratic potential by a general smooth function. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be smooth, assume that it has a unique global minimum at x_0 , that $\phi''(x_0) > 0$, and that $\phi(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$. Consider the integral

$$I_n(f) = \int_{\mathbb{R}} f(x) e^{-n\phi(x)} dx.$$

By the same reasoning as above, the main contribution comes from a neighborhood of the global minimum x_0 . Expanding ϕ around x_0 , we obtain

$$\phi\left(x_0 + \frac{s}{\sqrt{n}}\right) = \phi(x_0) + \frac{\phi''(x_0)}{2n} s^2 + \mathcal{O}(n^{-3/2}), \quad n \rightarrow \infty.$$

After the change of variables $x = x_0 + s/\sqrt{n}$, we obtain

$$I_n(f) = e^{-n\phi(x_0)} \sqrt{\frac{2\pi}{n\phi''(x_0)}} f(x_0) (1 + \mathcal{O}(n^{-1/2})), \quad n \rightarrow \infty.$$

More generally, suppose that ϕ has finitely many local minima, all of which are non-degenerate (i.e. $\phi'(x_0) = 0$ and $\phi''(x_0) > 0$). Then

$$I_n(f) = \sum_{x_0: \phi'(x_0)=0} e^{-n\phi(x_0)} \sqrt{\frac{2\pi}{n\phi''(x_0)}} f(x_0) (1 + \mathcal{O}(n^{-1/2})), \quad n \rightarrow \infty.$$

This is particularly useful when the global minimum is attained at several points.

Finally, note that the $\sqrt{n}$ -scaling arises from the assumption that $\phi''(x_0) > 0$. In degenerate cases where $\phi''(x_0) = 0$, the scaling changes and depends on the order of vanishing of $\phi'(x)$ at x_0 ; the argument can be adapted accordingly.

2. Complex integrals

The method becomes more interesting in the complex setting, where one considers integrals of the form

$$I_n = \oint_{\mathcal{C}} f(z) e^{-n\phi(z)} dz,$$

where $\mathcal{C}$ is a simple contour in the complex plane and f and ϕ are meromorphic functions.

Such integrals are analyzed in a way analogous to real Laplace-type integrals, but the complex setting introduces several subtleties that must be handled with care in each concrete situation. The basic idea is the following.

Since contour integrals of holomorphic functions are invariant under contour deformation, we try to deform the contour so that it passes through critical points of ϕ . Note that every critical point of ϕ is also a saddle point of $\operatorname{Re}\phi$, as a consequence of the Cauchy–Riemann equations.

After deforming the contour through a critical point, we further deform it so that it follows paths along which $\operatorname{Re}\phi$ decreases as fast as possible. At a non-degenerate saddle point there are two such directions of steepest descent (and two of steepest ascent). We therefore choose contours that pass through the saddle point and follow these directions.

A key observation is that, by the Cauchy–Riemann equations, the gradient of $\operatorname{Re}\phi$ is orthogonal to the level curves of $\operatorname{Im}\phi$. Consequently, the paths of steepest descent and ascent are contained in the level sets

$$\{z \in \mathbb{C} : \operatorname{Im}\phi(z) = \operatorname{Im}\phi(\omega)\},$$

where ω is a critical point of ϕ .

Summarizing, the strategy is as follows.

- (1) Identify the critical points ω of ϕ .
- (2) Determine the steepest descent contours through each critical point ω .
These curves lie in the set

$$\{z : \operatorname{Im}\phi(z) = \operatorname{Im}\phi(\omega)\}.$$

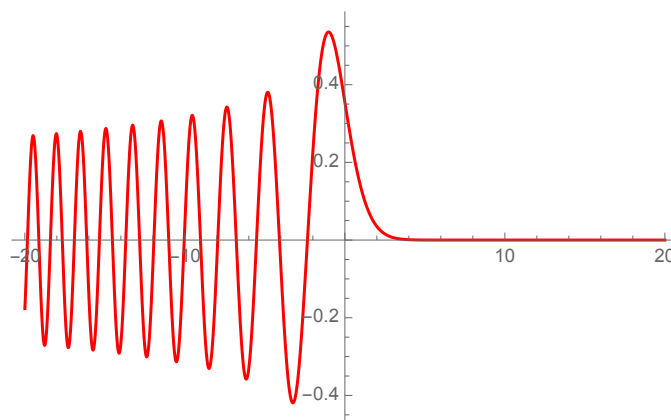


FIGURE 1. The Airy function $\text{Ai}(x)$. For $x < 0$ the function oscillates, while for $x > 0$ it decays exponentially.

- (3) Deform the original contour so that it passes through some of the critical points and follows steepest descent directions.
- (4) If possible, show that the main contribution to the integral comes from the dominant critical points, and analyze these contributions by introducing local coordinates near the saddle points.

This is a general guiding principle rather than a theorem, and it must be worked out in detail in each specific problem.

3. Solving the Airy equation

We work out an important and illustrative special case by characterizing the solutions of the second-order ordinary differential equation

$$(11) \quad y''(x) = xy(x).$$

This is called the *Airy equation*. It is the simplest example of a differential equation with a *turning point*: the qualitative nature of the solutions changes depending on the sign of x .

Indeed, for large positive x , the ratio $y''(x)/y(x)$ is positive, and we expect solutions to grow or decay exponentially. In contrast, for large negative x , the ratio $y''(x)/y(x)$ is negative, and we expect oscillatory behaviour. Thus, the solutions exhibit qualitatively different behaviour as $x \rightarrow +\infty$ and $x \rightarrow -\infty$.

We now construct solutions in integral form. Let γ_j , $j = 0, 1, 2$, be contours in the complex plane connecting $e^{(2j-1)\pi i/3}\infty$ to $e^{(2j+1)\pi i/3}\infty$, and define

$$(12) \quad y_j(x) = \frac{1}{2\pi i} \int_{\gamma_j} \exp\left(\frac{t^3}{3} - xt\right) dt, \quad j = 0, 1, 2.$$

A straightforward integration by parts shows that each y_j satisfies the Airy equation (11). Since the equation is second order, the three solutions must be linearly dependent; indeed, one checks that

$$y_0(x) + y_1(x) + y_2(x) = 0.$$

Any two of the functions y_0, y_1, y_2 form a basis of the solution space.

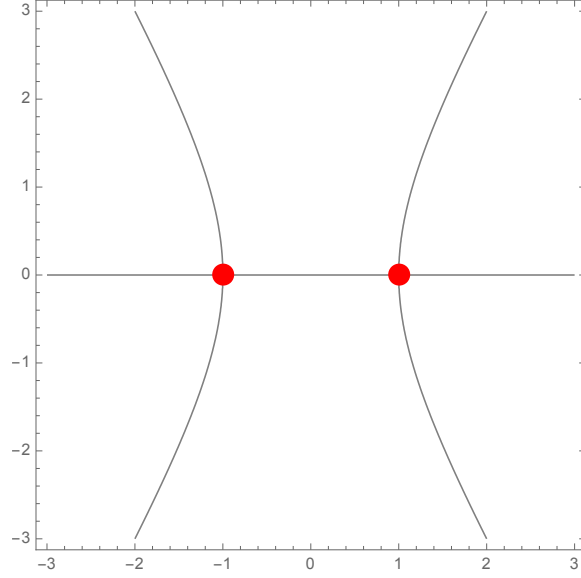


FIGURE 2. The saddle points ± 1 for the function $\phi(t) = t^3/3 - t$. The contours of steepest descent leaving from $t = +1$ have to stay in the first quadrant since $\text{Re}\phi(i\mathbb{R}) = 0$ and $(0, \infty)$ consists of steepest ascent paths leaving from $t = +1$.

THEOREM 10. Let y_0 be the solution to (11) given by (12) with contour γ_0 . Then

$$y_0(x) \sim \begin{cases} \frac{\exp(-\frac{2}{3}x^{3/2})}{2\sqrt{\pi}x^{1/4}}, & x \rightarrow +\infty, \\ \frac{\sin(\frac{2}{3}(-x)^{3/2} + \frac{\pi}{4})}{\sqrt{\pi}(-x)^{1/4}}, & x \rightarrow -\infty. \end{cases}$$

PROOF. We only carry out the full analysis for the asymptotic behaviour as $x \rightarrow +\infty$.

We rescale the integral (12) by setting $t = x^{1/2}s$:

$$(13) \quad y_0(x) = \frac{x^{1/2}}{2\pi i} \int_{\gamma_0} \exp\left(x^{3/2}\left(\frac{s^3}{3} - s\right)\right) ds = \frac{x^{1/2}}{2\pi i} \int_{\gamma_0} \exp\left(x^{3/2}\phi(s)\right) ds,$$

where

$$\phi(s) = \frac{s^3}{3} - s.$$

The critical points of ϕ are given by $\phi'(s) = s^2 - 1 = 0$, hence $s = \pm 1$. For $x > 0$, we deform the contour γ_0 so that it passes through $s = 1$ and follows steepest descent paths, while avoiding the other critical point $s = -1$. Moreover,

$$\phi(1) = -\frac{2}{3}, \quad \phi''(1) = 2.$$

We introduce a local variable near the saddle point $s = 1$ by setting

$$s = 1 + \frac{u}{\sqrt{2}x^{3/4}}.$$

Then

$$\phi(s) = -\frac{2}{3} - \frac{u^2}{2x^{3/2}} + \mathcal{O}(x^{-9/4}), \quad x \rightarrow \infty,$$

and therefore

$$x^{3/2}\phi(s) = -\frac{2}{3}x^{3/2} - \frac{u^2}{2} + \mathcal{O}(x^{-3/4}).$$

Substituting this into (13), we obtain

$$y_0(x) = \frac{x^{1/2}}{x^{3/4}2\pi\sqrt{2}} e^{-\frac{2}{3}x^{3/2}} \int_{-\infty}^{\infty} e^{-u^2/2} du \left(1 + \mathcal{O}(x^{-3/4})\right), \quad x \rightarrow +\infty.$$

Evaluating the Gaussian integral gives

$$y_0(x) = \frac{1}{2\sqrt{\pi}} x^{-1/4} e^{-\frac{2}{3}x^{3/2}} \left(1 + \mathcal{O}(x^{-3/4})\right), \quad x \rightarrow +\infty,$$

which proves the stated asymptotic formula.

The analysis for $x < 0$ proceeds in the same way. In this case, we rescale by $(-x)^{3/2}$, which again leads to the phase function

$$\phi(t) = \frac{t^3}{3} + t,$$

with two critical points at $\pm i$. One can deform the contour γ_0 so that it passes through these critical points and follows the paths of steepest descent.

Both saddle points now contribute equally, and one must perform a local analysis in a neighbourhood of each saddle point. This yields two complex exponential contributions, which together combine into a trigonometric function. We leave the remaining details to the reader. □

REMARK 11. *The solution y_1 does not decay exponentially as $x \rightarrow +\infty$; instead, it grows exponentially. This is because the contour defining y_1 passes through the other saddle point, which is dominant in that direction. Consequently, any linear combination of y_0 and y_1 will grow exponentially unless the contribution of y_1 vanishes.*

The solution y_0 plays a distinguished role and is called the *Airy function*, usually denoted by $\text{Ai}(x)$.

CHAPTER 3

The Aztec diamond

1. Determinantal structure of domino tilings

The uniform distribution on domino tilings of the Aztec diamond of size N gives rise to a determinantal point process with correlation kernel

$$(14) \quad \begin{aligned} K((2p + \varepsilon, y), (2q + \varepsilon', y')) &= -\mathbf{1}_{\{2p + \varepsilon < 2q + \varepsilon'\}} \frac{1}{2\pi i} \oint_{\gamma_z} (1+z)^{q-p+\varepsilon'-\varepsilon} \left(1 - \frac{1}{z}\right)^{-(q-p)} \frac{dz}{z^{y'-y+1}} \\ &\quad + \frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw (1+z)^{-p} \left(1 - \frac{1}{z}\right)^{-(N-p)} \\ &\quad \times (1+w)^q \left(1 - \frac{1}{w}\right)^{N-q} \frac{1}{w(w-z)} \frac{z^y}{w^{y'}}. \end{aligned}$$

where γ_z is any contour around $z = 0, 1$, but not $z = -1$, and γ_w goes around γ_z . Here $x = 0, \dots, N$ and $y \in \mathbb{Z}$ satisfy $y \geq -N + x$.

It is convenient to perform the change of variables

$$\begin{cases} x = x, \\ u = y + x. \end{cases}$$

This defines a determinantal point process on the set

$$\{-N, -N+1, \dots\}^{2N}.$$

A closer inspection of admissible path endpoints shows that we may in fact restrict to

$$\{-N, -N+1, \dots, -2, -1\}^{2N}.$$

In what follows we always set $\varepsilon = \varepsilon' = 0$, so that we consider vertical sections at even coordinates. This assumption is made only to minimize technicalities. Once the reader is familiar with the analysis below, it is a useful exercise to extend the arguments to vertical sections at odd coordinates.

(15)

$$\begin{aligned} K((2x, u), (2x', u')) &= -\mathbf{1}_{\{x < x'\}} \frac{1}{2\pi i} \oint_{|z|=1} (1+z)^{x'-x} (z-1)^{x'-x} \frac{dz}{z^{u'-u+1}} \\ &\quad + \frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw \frac{(1+w)^{x'} (w-1)^{N-x'}}{(1+z)^x (z-1)^{N-x}} \frac{1}{z(z-w)} \frac{z^{u+N}}{w^{u'+N}}, \end{aligned}$$

where γ_z is any contour around $z = 0, 1$, but not $z = -1$, and γ_w goes around γ_z .

2. The saddle points

In the asymptotic analysis as $N \rightarrow \infty$ we study the saddle points of the function

$$F(z) = F(z; \xi, \eta) = \xi \log(1+z) + (1-\xi) \log(z-1) - (\eta+1) \log z.$$

We ignore branch choices of the logarithm, since we ultimately analyze $e^{\pm NF(z)}$, and any factors arising from branch cuts cancel out.

A direct computation gives

$$F'(z) = \frac{1 + \eta + (1 - 2\xi)z - \eta z^2}{z - z^3}.$$

Hence the saddle points are the solutions of

$$1 + \eta + (1 - 2\xi)z - \eta z^2 = 0,$$

and are thus given by

$$\frac{1 - 2\xi \pm \sqrt{(1 - 2\xi)^2 + 4\eta(1 + \eta)}}{2\eta}.$$

The nature of the roots is determined by the discriminant

$$D(\xi, \eta) = (1 - 2\xi)^2 + 4\eta(1 + \eta) = 4 \left(\xi - \frac{1}{2} \right)^2 + 4 \left(\eta + \frac{1}{2} \right)^2 - 1.$$

Thus $D(\xi, \eta) = 0$ defines a circle in the (ξ, η) -plane with center $(\frac{1}{2}, -\frac{1}{2})$ and radius $\frac{1}{2}$. We will show that this circle coincides with the arctic circle of the Aztec diamond, separating the disordered region from the frozen corners. Before we come to that we mention a proposition, which is easy to check. First some notation.

Define

$$\mathcal{D} = \{(\xi, \eta) \in [0, 1] \times [0, -1] \mid D(\xi, \eta) < 0\},$$

and for $(\xi, \eta) \in \mathcal{D}$ define $\Omega(\xi, \eta)$ as

$$\frac{1 - 2\xi - i\sqrt{-D(\xi, \eta)}}{2\eta}.$$

PROPOSITION 12. *The map $(\xi, \eta) \mapsto \Omega(\xi, \eta)$ is a diffeomorphism between $\mathcal{D}$ and the upper half-plane $\{\text{Im} z > 0\}$. Moreover, it admits an extension to the boundary that maps the boundary of $\mathcal{D}$ bijectively onto the real line (including infinity), with the identification of the following points $\Omega(\frac{1}{2}, 0) = \infty$, $\Omega(1, -\frac{1}{2}) = 1$, $\Omega(\frac{1}{2}, -1) = 0$ and $\Omega(0, -\frac{1}{2}) = -1$.*

2.1. The frozen corners.

THEOREM 13. *Define the frozen region as*

$$F = ([0, 1] \times [-1, 0]) \setminus \overline{\mathcal{D}}.$$

Then

$$\lim_{N \rightarrow \infty} K_N([N\xi], [N\eta], [N\xi], [N\eta]) = \begin{cases} 0, & \text{if } \frac{1}{2} < \xi < 1, \\ 1, & \text{if } 0 < \xi < \frac{1}{2}. \end{cases}$$

PROOF. We will only prove this for the case that $(\xi, \eta) \in F$ and

$$\frac{1}{2} < \xi < 1, \quad -\frac{1}{2} < \eta < 0.$$

The proof for the other corners is similar.

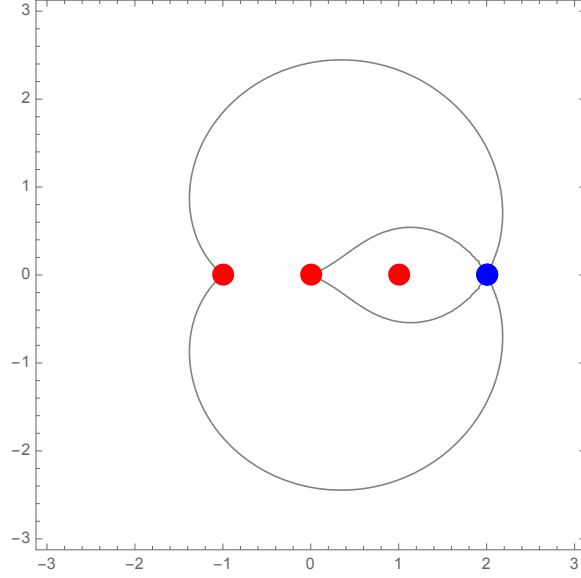


FIGURE 1. Contours of steepest ascent/descent leaving from the two real saddle points (blue dots). The red dots are the poles at $-1, 0$ and 1 .

We also ignore taking the integer part in the kernel (including the integer part is bookkeeping, and dropping it does not alter the heart of the matter). Then we get

$$K_N([N\xi], [N\eta]), ([N\xi], [N\eta]) = \frac{1}{(2\pi i)^2} \oint dz \oint dw \frac{e^{NF(w)}}{e^{NF(z)}} \frac{1}{z(z-w)},$$

where γ_z encloses γ_w and ± 1 and 0 .

The saddle points are both to the right of $+1$. We denote them by $1 < s_z < s_w$. By the behavior at the 1 and $+\infty$, and the fact that the $s_{w,z}$ are critical points, we find

$$\operatorname{Re} F(s_w) < \operatorname{Re} F(s_z).$$

We deform γ_z so that it passes through s_z and consists of the two steepest ascent contours leaving from s_z . These contours have to end at 0 (the only other option is that they end in another critical point or at $+\infty$. In both cases they would intersect the contours of steepest descent leaving from the other saddle point and this would violate the ordering). We deform γ_w so that it passes through s_w and follows contours of steepest descent leaving from s_w . These contours end at -1 , since ending at $+1$ would imply they cross the contours γ_z which would violate the ordering. See Figure 1.

By standard steepest descent argument arguments we find then that

$$K_N([N\xi], [N\eta]), ([N\xi], [N\eta]) = \mathcal{O}(N^{-1} e^{-N(\operatorname{Re} F(s_w) - \operatorname{Re} F(s_z))}), \quad N \rightarrow \infty.$$

and since $\operatorname{Re} F(s_w) < \operatorname{Re} F(s_z)$ this is exponentially decaying in N . $\square$

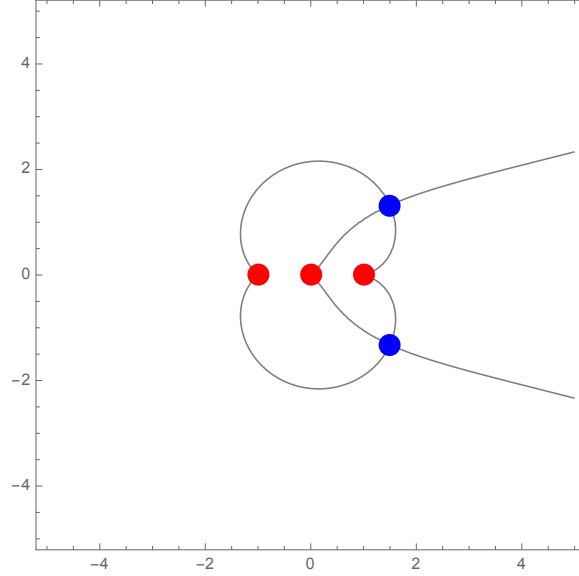


FIGURE 2. Contours of steepest ascent/descent leaving from the two complex conjugate saddle points (blue dots). The red dots are the poles at $-1, 0$ and 1 .

Since the kernel on the diagonal is the probability of having a point at that location, we see that in the frozen region the probability is either 0 or 1. There is no randomness in the paths.

2.2. The disordered region. In the disordered region we argue as follows. We recall that if $(\xi, \eta) \in \mathcal{D}$, then the saddle point equation has two complex conjugate solutions and the saddle point with positive imaginary part is denoted by $\Omega(\xi, \eta)$.

We first obtain the limiting density in the bulk.

THEOREM 14. *Let $(\xi, \eta) \in \mathcal{D}$. Then*

$$K_N([N\xi], [N\eta], [N\xi], [N\eta]) = \frac{1}{\pi} \arg \Omega(\xi, \eta) + \mathcal{O}(N^{-1/2}), \quad N \rightarrow \infty.$$

PROOF. There is a symmetry in the saddle points and in the contours of steepest descent/ascent. There are two contours of steepest ascent leaving from $\Omega(\xi, \eta)$: one ends at ∞ and the other connects to 0. The contours of steepest descent connect to ± 1 . By symmetry, the contours of steepest descent/ascent leaving from $\overline{\Omega(\xi, \eta)}$ are obtained by reflection with respect to the real axis. See Figure 2.

We now deform γ_z and γ_w to these steepest ascent/descent contours. In doing so, the resulting contours cross. Because of the factor $1/(w - z)$, this crossing produces a residue contribution. As a result we obtain a single integral term in

addition to a (deformed) double integral:

$$(16) \quad K_N([N\xi], [N\eta], ([N\xi], [N\eta])) = \frac{1}{2\pi i} \int_{-\Omega(\xi, \eta)}^{\Omega(\xi, \eta)} \frac{dw}{w} \\ + \frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw \frac{e^{NF(w)}}{e^{NF(z)}} \frac{1}{z(z-w)}.$$

After introducing local variables near $\Omega(\xi, \eta)$ (and its conjugate) in the double integral, one shows by standard steepest descent estimates that

$$\frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw \frac{e^{NF(w)}}{e^{NF(z)}} \frac{1}{w(w-z)} = \mathcal{O}(N^{-1/2}), \quad N \rightarrow \infty.$$

On the other hand, the single integral can be computed explicitly:

$$\frac{1}{2\pi i} \int_{-\Omega(\xi, \eta)}^{\Omega(\xi, \eta)} \frac{dw}{w} = \arg \Omega(\xi, \eta).$$

This finishes the proof. $\square$

Now that we have identified a non-trivial limiting density, the next question is the local randomness near a point in the disordered region. We zoom in and set

$$\begin{cases} x = [\xi N] + s, \\ u = [\eta N] + t, \\ x' = [\xi N] + s', \\ u' = [\eta N] + t', \end{cases} \quad s, t, s', t' \in \mathbb{Z}.$$

Then we have the following theorem.

THEOREM 15. *Let $(\xi, \eta) \in \mathcal{D}$. Then*

$$(17) \quad \lim_{N \rightarrow \infty} K_N((2x, u), (2x', u')) = -\mathbf{1}_{\{s < s'\}} \frac{1}{2\pi i} \oint_{|z|=1} (1+z)^{s'-s} (z-1)^{s'-s} \frac{dz}{z^{t'-t+1}} \\ + \frac{1}{2\pi i} \int_{\Omega(\xi, \eta)}^{\Omega(\xi, \eta)} (1+z)^{s'-s} (z-1)^{s'-s} \frac{dz}{z^{t'-t+1}}$$

PROOF. We analyze

$$(18) \quad K_N((2x, u), (2x', u')) = -\mathbf{1}_{\{s < s'\}} \frac{1}{2\pi i} \oint_{|z|=1} (1+z)^{s'-s} (z-1)^{s'-s} \frac{dz}{z^{t'-t+1}} \\ + \frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw \frac{(1+w)^{x'} (w-1)^{N-x'}}{(1+z)^x (z-1)^{N-x}} \frac{1}{w(w-z)} \frac{z^{u+N}}{w^{u'+N}}.$$

With the scaling $x = [\xi N] + s$, $u = [\eta N] + t$ (and similarly for x', u'), the second term can be written in the form

$$\frac{1}{(2\pi i)^2} \oint_{\gamma_z} dz \oint_{\gamma_w} dw \left[\text{prefactor independent of } N \right] \frac{e^{NF(w)}}{e^{NF(z)}} \frac{1}{w(w-z)}.$$

The single integral term does not depend on N . The analysis of the double integral is the same as in the proof of the limiting density: one deforms the contours to steepest descent/ascent paths through the complex conjugate saddle points. The

contour crossing yields the residue contribution, producing the stated single integral term, while the remaining double integral vanishes as $N \rightarrow \infty$ by steepest descent. Keeping track of the extra prefactors yields the claimed limit. $\square$

2.3. Asymptotics near the arctic circle. We now study the asymptotic behavior of the kernel near the boundary of the domain $\mathcal{D}$, commonly referred to as the *arctic circle*. At such boundary points, two saddle points of the phase function coalesce, leading to a change in scaling behavior and to universal Airy-type limits.

To make this explicit, we focus on the distinguished boundary point

$$P = \left(\frac{1}{4}(2 + \sqrt{2}), -\frac{1}{4}(2 - \sqrt{2}) \right).$$

For this choice of P , the phase function $F(\cdot; P)$ has a *double critical point* at

$$\Omega(P) = 1 + \sqrt{2},$$

that is,

$$F'(\Omega(P); P) = F''(\Omega(P); P) = 0, \quad F'''(\Omega(P); P) \neq 0.$$

Since the first two derivatives vanish at $\Omega(P)$, the leading nontrivial term in the Taylor expansion of the phase is cubic. Expanding around the saddle point and multiplying by N , we obtain

$$NF(\Omega(P) + uN^{-1/3}; P) = NF(\Omega(P); P) + \frac{u^3}{6}F'''(\Omega(P); P) + \mathcal{O}(N^{-1/3}).$$

The natural scaling $u = O(1)$ therefore corresponds to microscopic fluctuations of order $N^{-1/3}$ around the saddle.

It is convenient to normalize the cubic coefficient. After the rescaling

$$u \mapsto \frac{2^{1/3}u}{(F'''(\Omega(P); P))^{1/3}},$$

the expansion takes the canonical form

$$NF\left(\Omega(P) + \frac{2^{1/3}u}{(F'''(\Omega(P); P))^{1/3}N^{1/3}}; P\right) = NF(\Omega(P); P) + \frac{u^3}{3} + \mathcal{O}(N^{-1/3}),$$

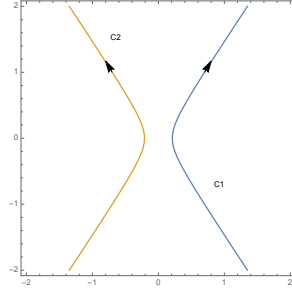
which already suggests the appearance of Airy-type integrals.

To capture correlations near the arctic circle, we introduce local coordinates around the point P . There are two distinguished directions:

- the *normal direction* $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, which must be scaled on the $N^{-1/3}$ level;
- the *tangential direction* $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, which turns out to scale on the finer $N^{-2/3}$ level.

With appropriate constants $c_0, c_1, c_2 \in \mathbb{R}$, one can arrange that the combined scaling

$$z = \Omega(P) + \frac{c_0 u}{N^{1/3}}, \quad P \mapsto P + c_1 s \begin{pmatrix} 1 \\ -1 \end{pmatrix} N^{-1/3} + c_2 t \begin{pmatrix} 1 \\ 1 \end{pmatrix} N^{-2/3}$$

FIGURE 3. The contours γ_1 on the right and γ_2 on the left.

leads to the expansion

$$\begin{aligned} NF\left(\Omega(P) + \frac{c_0 u}{N^{1/3}}; P + sc_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} N^{-1/3} + tc_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} N^{-2/3}\right) \\ = NF(\Omega(P); P) + \frac{u^3}{3} + \frac{s^2 u}{2} + tu + \mathcal{O}(N^{-1/3}). \end{aligned}$$

This is the universal cubic–quadratic–linear structure characteristic of edge scaling limits.

As a consequence of the above asymptotics, the rescaled kernel converges to an Airy-type kernel.

THEOREM 16. *There exists constants $c_0, c_1, c_2 \in \mathbb{R}$ such that*

$$\begin{aligned} (19) \quad \lim_{N \rightarrow \infty} \frac{c_0}{N^{1/3}} K_N \left(NP + c_1 s_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} N^{2/3} + c_2 t_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} N^{1/3}, \right. \\ \left. NP + c_1 s_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} N^{2/3} + c_2 t_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} N^{1/3} \right) \\ = -\frac{1}{2\pi i} \mathbf{1}_{s_1 < s_2} \int_{i\mathbb{R}} e^{(s_2 - s_1)u^2/2 + (t_2 - t_1)u} du \\ + \frac{1}{(2\pi i)^2} \int_{\gamma_2} \int_{\gamma_1} \frac{e^{u^3/3 + s_2 u^2/2 + t_2 u}}{e^{v^3/3 + s_1 v^2/2 + t_1 v}} \frac{du dv}{u - v}, \end{aligned}$$

where the contours γ_1 and γ_2 are as in Figure 3

PROOF. The contours γ_z and γ_w are deformed so that they pass through the double saddle point $\Omega(P) = 1 + \sqrt{2}$. The contour γ_w follows a path of steepest descent, while γ_z follows a path of steepest ascent with respect to $\operatorname{Re} F(z; P)$. See also Figure 4.

Since $\Omega(P)$ is a double zero of F'' , the steepest descent/ascent directions emerge at angles $\pm\pi/3$ and $2\pi/3$ relative to the real axis. After performing the above changes of variables and taking the limit $N \rightarrow \infty$, the stated expression follows. $\square$

REMARK 17 (Extended Airy kernel form). *The limiting kernel obtained above can be identified, up to a harmless gauge transformation, with the extended Airy kernel. By a gauge transformation we mean conjugation of the kernel by explicit*

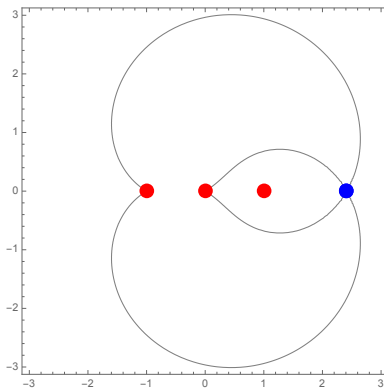


FIGURE 4. Steepest descent/ascent contours in case of coalescing saddle points

exponential factors in the variables (s, t) , which does not affect any determinantal correlation functions.

One convenient way to see this is to use the identity

$$\frac{1}{u-v} = \int_0^\infty e^{-\lambda(u-v)} d\lambda, \quad \operatorname{Re}(u-v) > 0,$$

which allows us to factor the double contour integral into a product of two single integrals. After this step, the u - and v -integrals can be identified with (contour) integral representations of the Airy function.

The remaining Gaussian term supported on $\{s_1 < s_2\}$ corresponds to the heat-kernel part of the extended Airy kernel. Altogether, and after an appropriate change of variables and conjugation, the limit kernel takes the standard extended Airy kernel form

$$K_{\text{extAi}}(\tau_1, x_1; \tau_2, x_2) = -\mathbf{1}_{\tau_1 < \tau_2} p_{\tau_2 - \tau_1}(x_2 - x_1) + \int_0^\infty e^{-\lambda(\tau_2 - \tau_1)} \operatorname{Ai}(x_1 + \lambda) \operatorname{Ai}(x_2 + \lambda) d\lambda,$$

where $p_t(x) = (2\pi t)^{-1/2} e^{-x^2/(2t)}$ denotes the one-dimensional heat kernel.

2.4. Near the turning points.

CHAPTER 4

GUE

We discuss some famous results about the Gaussian Unitary Ensemble. One goal of the course is to develop techniques for proving these results.

1. Gaussian Unitary Ensemble

DEFINITION 18. *The Gaussian Unitary Ensemble (GUE) is the probability measure on the space of $n \times n$ Hermitian matrices such that the entries*

- M_{ij} for $1 \leq i \leq j \leq n$ are independent,
- $M_{ij} \sim \mathcal{N}_{\mathbb{C}}(0, 1/n)$ for $1 \leq i < j \leq n$,
- $M_{jj} \sim \mathcal{N}_{\mathbb{R}}(0, 1/n)$ for $1 \leq j \leq n$.

The GUE has a symmetry that is very important.

LEMMA 19. *The Gaussian Unitary Ensemble is invariant under conjugation with a unitary matrix, that is, the law of M is the same as the law of U^*MU for any $n \times n$ unitary matrix U .*

PROOF. (sketch) This is a consequence of the fact that a sum of independent Gaussians is Gaussian and the fact that the columns and rows of U and U^* are orthogonal unit vectors. $\square$

PROPOSITION 20. *The marginal distribution on the eigenvalues $\lambda_1, \dots, \lambda_n$ of a GUE matrix is given by*

$$\frac{1}{Z_n} \prod_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)^2 e^{-\frac{n}{2} \sum_{j=1}^n \lambda_j^2} \prod_{j=1}^n d\lambda_j$$

where Z_n is a normalizing constant.

PROOF. (sketch) This is a consequence of Weyl's integration theorem. It can also be computed using the jacobian of the map $M \mapsto (\Lambda, U)$ where Λ is a diagonal matrix of ordered eigenvalues and U is the unitary matrix such that $M = U\Lambda U^*$. This is a tedious computation. $\square$

We see that the jpdf for the eigenvalues vanishes when two eigenvalues coincide. The probability of clustering eigenvalues is very small and the eigenvalues tend to repel each other. This effect by itself will push the eigenvalues far out, but the Gaussian density prevents the eigenvalues from escaping to infinity. Ultimately, the eigenvalues will try to find a balance between these two effects.

⁰The reason for taking the variance equal to $1/n$ will be clear later.

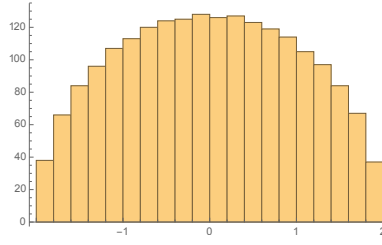


FIGURE 1. The figure shows a histogram for a 2000×2000 GUE matrix. The semi-circle is clearly visible.

As we have seen in the chapter on biorthogonal ensembles, this is an example of a so-called orthogonal polynomials ensembles. Let p_k be the unique polynomial of degree k and positive leading coefficient such that

$$(20) \quad \int p_k(x)p_j(x)e^{-nx^2/2}dx = \begin{cases} 1, & j = k \\ 0, & j \neq k. \end{cases}$$

Then it follows from Theorem ?? that the eigenvalues of the GUE matrix form a determinantal point process with kernel

$$K_n(x, y) = e^{-nx^2/2} \sum_{j=0}^{n-1} p_j(x)p_j(y)$$

with the Lebesgue measure as the reference measure.

The orthogonal polynomials p_n are rescaled and normalized versions of the so-called Hermite polynomials. These are classical orthogonal polynomials and their properties have been well-studied. What we will see is that such properties can be used to prove various claims on the limiting behavior of the eigenvalues of a GUE matrix.

2. Some limit results

We now state a few results on the limiting behavior of the eigenvalues when the size of the matrix $n \rightarrow \infty$.

2.1. Semi-Circle law. The jpdf for the eigenvalues can be viewed as as a Boltzman-Gibbs type of measure

$$e^{-n^2\mathcal{E}(\lambda_1, \dots, \lambda_n)} d\lambda_1 \dots d\lambda_n,$$

where

$$\mathcal{E}(\lambda_1, \dots, \lambda_n) = -\frac{1}{n^2} \sum_{i \neq j} \log |\lambda_i - \lambda_j| + \frac{1}{n} \sum_{j=1}^n \frac{\lambda_j^2}{2}.$$

Here we see a first important principle. As $n \rightarrow \infty$, the eigenvalues will be close to the configuration that minimizes the energy function $\mathcal{E}$. In the limit as $n \rightarrow \infty$, this will imply that the distribution of eigenvalues will be given by

$$\frac{1}{n} \sum_{j=1}^n \delta_{\lambda_j} \rightarrow \frac{1}{2\pi} \sqrt{4 - \lambda^2} d\lambda, \quad \text{on } [-2, 2].$$

The repulsion between the eigenvalues is very strong.

THEOREM 21. *Let $A \subset \mathbb{R}$ be a bounded measurable set. Then*

$$\frac{1}{n} \mathbb{E}[N(A)] = \frac{1}{n} \int_A K_n(x, x) d\mu(x) \rightarrow \frac{1}{2\pi} \int_A \sqrt{2 - x^2} dx$$

We will prove this result in this chapter using properties of the correlation kernel for the GUE.

We will see later that in fact a much stronger result is true and that we will have convergence with probability 1. That is, as $n \rightarrow \infty$,

$$\frac{1}{n} (N(A) - \mathbb{E}[N(A)]) \rightarrow 0$$

with probability 1. But this statement requires more knowledge on the structure of biorthogonal ensembles.

2.2. Bulk universality. The repulsion ensures that the eigenvalues are nicely distributed on the interval $[-2, 2]$. Since we have n points, we expect the distance between neighboring eigenvalues to be of order $1/n$. This is indeed true near any bulk point $x^* \in (-2, 2)$.

The celebrated sine universality, says that if we blow up the process near x_* so that the scaled eigenvalues are at distance ~ 1 , then the eigenvalues will converge to a new point process called the sine process. That is, consider the random points

$$x_j = (\lambda_j - x_*) \frac{n\sqrt{4 - x_*^2}}{2\pi}.$$

Then the random point process defined by the $\{x_j\}_j$ converges to the determinantal point process with kernel

$$K_{\text{sine}}(x, y) = \frac{\sin \pi(x - y)}{\pi(x - y)}.$$

This follows from the following result

THEOREM 22.

$$\frac{2\pi}{n\sqrt{4 - x_*^2}} K_n \left(x_* + \frac{2\pi x}{n\sqrt{4 - x_*^2}}, x_* + \frac{2\pi y}{n\sqrt{4 - x_*^2}} \right) \rightarrow K_{\text{sine}}(x, y) \frac{G(x)}{G(y)}$$

uniformly for x, y in compact subsets of $\mathbb{R}$.

Indeed, once one proves this result, then the by the continuity of the Fredholm determinant one can prove convergence of the finite dimensional distribution (it is in this sense that we mean convergence). This in particular, proves the following result on gap probabilities Let $I \subset \mathbb{R}$ be an interval. Then,

$$\mathbb{P} \left(\text{No eigenvalues in the interval } x_* + \frac{2\pi I}{n\sqrt{4 - x_*^2}} \right) \rightarrow \det(I - K_{\text{sine}})_{L_2(I)}$$

where we recall for completeness that

$$\begin{aligned} \det(I - K_{\text{sine}})_{L_2(I)} \\ = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \int_I \cdots \int_I \det (K_{\text{sine}}(x_i, x_j))_{i,j=1}^k dx_1 \cdots dx_k \end{aligned}$$

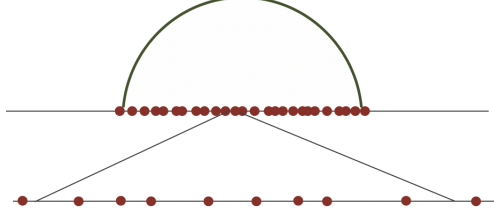


FIGURE 2. Scaling limit in the bulk

The right-hand side of (2.2) does not depend on x^* and holds for any $x_* \in (-2, 2)$. (but not at ± 2).

The universality conjecture (now proved in many different situations) says that the above law will hold for a wide class of random matrix models.

2.3. Edge Universality. At the edge we have the following result.
Scale the eigenvalues as

$$x_j = c(\lambda_j - 2)n^{2/3}.$$

Where c is some special numeric constant. Then, the finite dimensional distributions for the point process converge to that of the Airy process, which is the determinantal point process with kernel

$$K_{\text{Airy}}(x, y) = \frac{\text{Ai}(x)\text{Ai}'(y) - \text{Ai}(y)\text{Ai}'(x)}{x - y}$$

By continuity of the Fredholm determinant this follows from the following result

THEOREM 23. *As $n \rightarrow \infty$, we have*

$$\frac{e^{n^{1/3}(v-u)}}{n^{2/3}} K_n\left(2 + \frac{u}{cn^{2/3}}, 2 + \frac{v}{cn^{2/3}}\right) \rightarrow K_{\text{Airy}}(u, v)$$

uniformly for x, y from compact subsets of $\mathbb{R}$.

The next type of universality is on the distribution of the large eigenvalue.

Let λ_n be the largest eigenvalue of a GUE matrix. Then the location will be approximately at the edge of the support of the semi-circle law. That is $\lambda_n \approx 2$. It turns out it fluctuates at the order $n^{-2/3}$. This means that the fluctuations of $(\lambda_n - 2)n^{2/3}$ are of order 1.

There exists a numeric constant $c > 0$ such that

$$\mathbb{P}\left(c(\lambda_n - 2)n^{2/3} \leq s\right) \rightarrow \det(I - K_{\text{Airy}})_{L_2(s, \infty)}$$

where we recall for completeness that

$$\begin{aligned} \det(I - K_{\text{Airy}})_{L_2(s, \infty)} \\ = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \int_s^{\infty} \cdots \int_s^{\infty} \det(K_{\text{Airy}}(x_i, x_j))_{i,j=1}^k dx_1 \cdots dx_k \end{aligned}$$

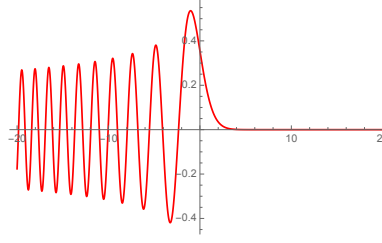


FIGURE 3. The Airy function is the unique solution to the equation

$$y''(s) = sy(s) \text{ such that } y(s) = \frac{e^{-\frac{2}{3}s^2}}{2\sqrt{\pi}s^{\frac{1}{4}}}(1 + o(1)), \text{ as } s \rightarrow +\infty.$$

Now it is important that this result does NOT follow from the convergence of the finite dimensional distribution. Indeed, the interval $[s, \infty)$ is unbounded.

This is in fact a subtle fact. The convergence of the finite dimensional distribution is not strong enough to imply convergence of the distribution of the large eigenvalue. In fact, without further arguments, it could be true that the largest eigenvalues runs off to infinity (which will not happen). We thus have to prove more (which we will do).

3. Integral representations

Hermite polynomials have the following two different integral representations.

LEMMA 24. *Set*

$$p_k(x) = \frac{n}{2\pi i} \int_{\gamma_1} e^{\frac{n(z-x)^2}{2}} z^k dz$$

where γ_1 is a contour connection $-i\infty$ to $+i\infty$, and

$$q_k(x) = \frac{1}{2\pi i} \oint_{\gamma_2} e^{-n(\frac{z^2}{2} - xz)} \frac{dz}{z^{k+1}}$$

where γ_2 is counterclockwise oriented contour around the origin.¹

Then q_j and p_k are polynomials of degree j and k and

$$\int_{-\infty}^{\infty} p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx = \delta_{jk}.$$

PROOF. Take γ_1 and γ_2 so that γ_1 lies at the right of γ_2 . Then

$$\begin{aligned} (21) \quad & \int_{-\infty}^{\infty} p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx \\ &= \int_{-\infty}^0 p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx + \int_0^{\infty} p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx \end{aligned}$$

First we deform γ_1 to $\tilde{\gamma}_1$ lies at the left of γ_2 . By switching integrals and using that

$$\int_{-\infty}^0 n e^{n(z-w)x} = \frac{1}{z-w}$$

¹Note that for each k there exists a constant c_k such that $p_k = c_k q_k$.

, we find

$$\int_{-\infty}^0 p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx = \frac{1}{(2\pi i)^2} \int_{\tilde{\gamma}_1} \oint_{\gamma_2} \frac{e^{n(\frac{z^2}{2}-xz)} z^k}{e^{n(\frac{w^2}{2}-xw)} w^{j+1}} \frac{dw dz}{z-w}$$

Then we deform γ_1 to $\hat{\gamma}_1$ so that it lies at the right of γ_2 . By switching integrals and using that

$$\int_0^\infty n e^{n(z-w)x} = -\frac{1}{z-w}$$

, we find

$$\int_0^\infty p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx = -\frac{1}{(2\pi i)^2} \int_{\hat{\gamma}_1} \oint_{\gamma_2} \frac{e^{n(\frac{z^2}{2}-xz)} z^k}{e^{n(\frac{w^2}{2}-xw)} w^{j+1}} \frac{dw dz}{z-w}$$

The minus sign can be removed by reversing the orientation. Then adding the two integrals gives.

$$\int_{-\infty}^\infty p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx = \frac{1}{(2\pi i)^2} \oint_{\gamma_1} \oint_{\gamma_2} \frac{e^{n(\frac{z^2}{2}-xz)} z^k}{e^{n(\frac{w^2}{2}-xw)} w^{j+1}} \frac{dw dz}{z-w}$$

where γ_1 is now a clockwise oriented contour that goes around γ_2 . By residue calculus and the fact that that integral over γ_1 only has a pole at $z = w$, we thus find

$$\int_{-\infty}^\infty p_k(x) q_j(x) e^{-\frac{nx^2}{2}} dx = \frac{1}{2\pi i} \oint_{\gamma_2} \frac{dz}{z^{j-k+1}},$$

This proves the statement. $\square$

COROLLARY 25. *The GUE is a determinantal point process with kernel*

$$K_n(x, y) = \frac{n}{(2\pi i)^2} \int_{\gamma_1} \oint_{\gamma_2} \frac{e^{n(\frac{z^2}{2}-xz)} z^n}{e^{n(\frac{w^2}{2}-xw)} w^n} \frac{dw dz}{z-w}$$

where γ_2 is counterclockwise oriented contour around the origin and γ_1 is a contour connection $-i\infty$ to $+i\infty$ and is at the right of γ_2 .

PROOF. Since $p_k(x)$ and $q_k(x)$ are the Hermite polynomial of degree k but with different normalization, we can write

$$K_n(x, y) = e^{-\frac{nx^2}{2}} \sum_{k=0}^{n-1} p_k(x) q_k(y).$$

Now the statement easily follows by using the integral representations for q_k and p_k and using

$$\sum_{k=0}^{n-1} \frac{z^k}{w^{k+1}} = \frac{1 - \frac{z^n}{w^n}}{w - z}.$$

Indeed this leads to

$$K_n(x, y) = \frac{n}{(2\pi i)^2} \int_{\gamma_1} \oint_{\gamma_2} \frac{e^{n(\frac{z^2}{2}-xz)}}{e^{n(\frac{w^2}{2}-xw)}} \left(1 - \frac{z^n}{w^n}\right) \frac{dw dz}{w - z}.$$

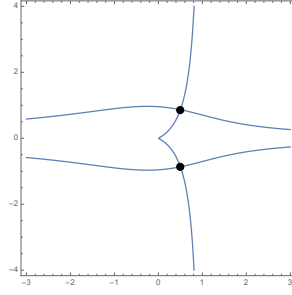


FIGURE 4. The picture shows the location of the saddle points and part of the level set where the imaginary point is constant. The paths emanating from the saddle points are paths of steepest descent/ascent for $\text{Re}F$

If we take γ_1 so that it lies at the right of γ_2 we see that γ_2 only has a pole at $w = 0$ for the term w^n . By splitting the integral into two, we thus arrive at the statement. $\square$

4. Steepest descent analysis for q_n

Let us first look how one can compute the asymptotic behavior for the polynomial

$$q_n(x) = \frac{1}{2\pi i} \oint e^{-n\left(\frac{z^2}{2} - xz\right)} \frac{dz}{z^{n+1}}$$

Note that we can write the integral as

$$q_n(x) = \frac{1}{2\pi i} \oint_{\gamma_2} e^{-nF(z)} \frac{dz}{z}$$

where

$$F(z) = \frac{z^2}{2} - xz + \log z.$$

The first point of the method is that we expect that the main contribution of this integral comes from points where $\text{Re}F(z)$ is minimal. Since $\text{Re}F(z)$ is harmonic, there are no local minima. There are however saddle points. If we could change the contour of integration so that it passes through certain saddle points but then continues in directions where $\text{Re}F(z)$ is increasing, then we expect the main contribution to come from those saddle points.

Let us find the critical points of $F(z)$ (these are the saddle points $\text{Re}F(z)$). Since

$$F'(z) = z - x + 1/z,$$

these point are given by

$$z = \frac{x \pm \sqrt{x^2 - 4}}{2}.$$

We thus see that if $-2 < x < 2$ then we have two complex conjugate solutions, and if $|x| > 2$ we have two real critical points. If $x = \pm 2$ then these critical points merge into a double critical point at $z = \pm 1$.

4.1. Regime $-2 < x < 2$. We want to deform the contour so that it goes through the points

$$z_s = \frac{x + i\sqrt{4-x^2}}{2} \quad \text{and} \quad \bar{z}_s = \frac{x - i\sqrt{4-x^2}}{2}.$$

But we also want that $\operatorname{Re}F(z)$ is minimal these points when z is on the contour γ_1 . By the Cauchy Riemann equations, the real part is increasing or decreasing most when the imaginary part remains constant. We therefore want the contour to be part of the level set

$$\{z \in \mathbb{C} \mid \operatorname{Im}F(z) = \operatorname{Im}F(z_s)\}.$$

(Note that $\operatorname{Im}F(z_s) = \operatorname{Im}F(\bar{z}_s)$). Observe that $\operatorname{Re}F(z)$ is increasing when we leave z_s (or $\bar{z}_s$) and continue to the parts that go to $\pm\infty$ (the other paths will play a role later). We thus deform the contour γ_2 so that it is exactly the union of these curves. These curves are *paths of steepest ascent*.

Now that we know that the $\operatorname{Re}F(z)$ is minimal as z_s and $\bar{z}_s$ we can throw away most parts of the contour, since their contributions will be exponentially small, and focus only on neighborhoods of the critical points. That is, we introduce variables

$$z = z_s + \frac{t}{n^{1/2}(F''(z_s))^{1/2}}$$

and

$$z = \bar{z}_s + \frac{t}{n^{1/2}(F''(\bar{z}_s))^{1/2}}$$

Care should be taken when we decide the square root. Note that $F''(z) = 1 - 1/z^2$ and since $|z_s| = 1$, we have $F''(z) = 1 - \bar{z}_s^2$. This means that $\operatorname{Re}F''(z_s) > 0$. We take the square root such that $\operatorname{Re}(F''(z_s))^{1/2} > 0$. Similarly, we take $\operatorname{Re}(F''(\bar{z}_s))^{1/2} > 0$. Then

$$\begin{aligned} (22) \quad & \frac{1}{2\pi i} \oint_{\gamma_2} e^{-nF(z)} \frac{dz}{z^1} \\ &= -\frac{1}{2\pi i(nF''(z_s))^{1/2}} \frac{e^{-nF(z_s)}}{z_s} \int_{-\infty}^{\infty} e^{-t^2/2} dt + \frac{e^{-nF(\bar{z}_s)}}{2\pi i(nF''(\bar{z}_s))^{1/2}} \frac{1}{\bar{z}_s} \int_{-\infty}^{\infty} e^{-t^2/2} dt \\ & \quad + \text{smaller order terms} \\ &= -\frac{e^{-nF(z_s)}}{i(2\pi nF''(z_s))^{1/2}} \frac{1}{z_s} + \frac{e^{-nF(\bar{z}_s)}}{i(2\pi nF''(\bar{z}_s))^{1/2}} \frac{1}{\bar{z}_s} + \text{smaller order terms} \end{aligned}$$

4.2. Regime $|x| > 2$. In the regime $|x| > 2$ we have two real critical points

$$z_1 = \frac{x - \sqrt{x^2 - 4}}{2} \quad \text{and} \quad z_2 = \frac{x + \sqrt{x^2 - 4}}{2}.$$

If $x > 0$ then both $0 < z_1 < z_2$ and if $x < 0$ then $z_2 < z_1 < 0$. We will only discuss the case $x > 0$.

In that case it is not hard to see that $\operatorname{Re}F(z_1) > \operatorname{Re}F(z_2)$. We deform the contour γ_2 now so that it continuous along the part of the level set

$$\{z \mid \operatorname{Im}F(z) = \operatorname{Im}F(z_2)\},$$

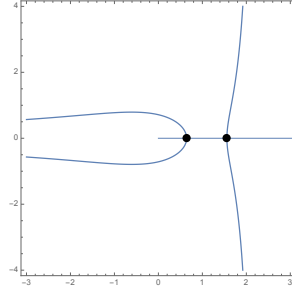


FIGURE 5. The picture shows the location of the saddle points and part of the level set where the imaginary point is constant. The paths emanating from the saddle points are paths of steepest descent/ascent

that are in the upper and lower half planes. Compared to the regime $-2 < x < 2$ we only have one critical point and the main contribution comes from a neighborhood around that point. After introducing local variable, we then find

$$\frac{1}{2\pi i} \oint_{\gamma_2} e^{-nF(z)} \frac{dz}{z} = \frac{e^{-nF(z_2)}}{z_2 \sqrt{2\pi n |F''(z_2)|}} + \text{lower order terms}$$

5. Steepest descent analysis for $\tilde{p}_n$

Now let us consider a similar analysis for

$$\tilde{p}_n(x) = e^{-\frac{nx^2}{2}} p_n(x) = \frac{1}{2\pi i} \int_{\gamma_1} e^{n\left(\frac{z^2}{2} - xz\right)} z^n dz.$$

Note that we can write the integral as

$$\tilde{p}_n(x) = \frac{1}{2\pi i} \int_{\gamma_1} e^{nF(z)} dz$$

where, again,

$$F(z) = \frac{z^2}{2} - xz + \log z.$$

We see that the only thing that changes is the sign in front of F . In other words, we want to deform γ_1 so that it passes through the critical points, but now we want to follow the parts of the level sets of $\text{Im}F(z)$ where $\text{Re}F(z)$ is strictly decreasing.

5.1. Regime $-2 < x < 2$. We want to deform the contour so that it goes through the points

$$z_s = \frac{x + i\sqrt{4-x^2}}{2} \quad \text{and} \quad \bar{z}_s = \frac{x - i\sqrt{4-x^2}}{2}.$$

We now argue exactly as before, but take the other parts of the level set

$$\{z \mid \text{Im}F(z) = \text{Im}F(z_s)\}.$$

That is, we take the curves that leave from z_s and $\bar{z}_s$ and end up in the origin and $\pm i\infty$. These curves are *paths of steepest descent* for $\text{Re}F(z)$. After that we introduce the local variables

$$z = z_s + \frac{it}{n^{1/2}(F''(z_s))^{1/2}}$$

and

$$z = \bar{z}_s + \frac{it}{n^{1/2}(F''(\bar{z}_s))^{1/2}}$$

(note that i in front of t .) We leave the details to the reader.

5.2. Regime $|x| > 2$. In the regime $|x| > 2$ we have two real critical points

$$z_1 = \frac{x - \sqrt{x^2 - 4}}{2} \quad \text{and} \quad z_2 = \frac{x + \sqrt{x^2 - 4}}{2}.$$

again, we will only discuss the case $x > 0$, which gives $\text{Re}F(z_1) > \text{Re}F(z_2)$. We deform the contour γ_1 now so that it continuous along the part of the level

$$\{z \mid \text{Im}F(z) = \text{Im}F(z_1)\},$$

that are in the upper and lower half planes. In other words, in contrast to the analysis for q_n , we now choose z_1 instead of z_2 . After introducing local variable, we then find

$$\frac{1}{2\pi i} \oint_{\gamma_1} e^{nF(z)} dz = \frac{e^{nF(z_1)}}{\sqrt{2\pi n|F''(z_1)|}} + \text{lower order terms}$$

6. Steepest descent analysis for $K_n(x, x)$

Now we come to the steepest descent analysis for the diagonal of the correlation kernel

$$K_n(x, x) = \frac{n}{(2\pi i)^2} \int_{\gamma_1} dz \oint_{\gamma_2} dw \frac{e^{n(F(z) - F(w))}}{z - w}$$

6.1. Regime $-2 < x < 2$. In the regime $-2 < x < 2$ we want to deform γ_1 and γ_2 to parts of the level set

$$\{z \mid \text{Im}F(z) = \text{Im}F(z_s)\}.$$

such that γ_1 has minima at $z = z_s$ and $z = \bar{z}_s$ and γ_2 has maxima at these points. This all seems to be reducing the analysis to individual integrals, but there is a catch: there is a pole at $z = w$.

We first deform γ_2 and put γ_1 at a location far away from γ_2 . Then we deform γ_1 to a steepest descent path leaving from the saddle points. Naturally γ_1 will now intersect γ_2 at these saddle points. In this deformation of γ_2 we thus pick up a residue for every $z \in \gamma_1$ that lies at the right of the steepest descent parts of the level set.

We thus find

$$K_n(x, x) = \frac{n}{(2\pi i)^2} \int_{\gamma_1} dz \oint_{\gamma_2} dw \frac{e^{n(F(z) - F(w))}}{z - w} + \frac{n}{2\pi i} \int_{\bar{z}_s}^{z_s} dz.$$

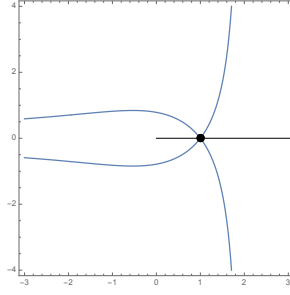
After introducing local variables, we can see that

$$\frac{n}{(2\pi i)^2} \int_{\gamma_1} dw \oint_{\gamma_2} dz \frac{e^{n(F(z) - F(w))}}{w - z} = \mathcal{O}(n^{1/2}).$$

The single integral reduces to

$$\frac{n}{2\pi i} \int_{\bar{z}_s}^{z_s} dz = \frac{n}{\pi} \text{Im}z_s = \frac{n\sqrt{4 - x^2}}{2\pi}.$$

¹Work this out for yourself. It is a good exercise!



After division by n this is exactly the semi-circle law!

6.2. Regime $|x| > 2$. The regime $|x| > 2$ is less cumbersome, as the steepest descent paths from z_2 does not intersect the steepest ascent paths from z_1 . The result is that we can prove that, for some constant $c > 0$,

$$K_n(x, x) = ce^{n(F(z_2) - F(z_1))} + \text{smaller order terms.}$$

Since $F(z_2) < F(z_1)$ we get exponential decay of the density.

7. Sine kernel asymptotics

Now we are ready for the sine kernel asymptotics. This is almost the same as the asymptotic for $K_n(x, x)$ since the integral changes only in a minor way

$$\frac{1}{n\rho(x_*)} \tilde{K}_n\left(x_* + \frac{x}{n\rho(x_*)}, x_* + \frac{y}{n\rho(x_*)}\right) = \frac{1}{\rho(x_*)(2\pi i)^2} \int_{\gamma_1} dz \oint_{\gamma_2} dw \frac{e^{n(F(z) - F(w))}}{z - w} e^{(wy - zx)/\rho(x_*)}$$

where F is take with respect to x_* .

Arguing as before, deforming contours as before, shows that the main contribution again comes from the single integral that now takes the form

$$\begin{aligned} \frac{1}{2\pi i \rho(x_*)} \int_{\bar{z}_s}^{z_s} e^{(y-x)z/\rho(x_*)} dz &= \frac{1}{2\pi i(y-x)} \left(e^{(y-x)z_s/\rho(x_*)} - e^{(y-x)\bar{z}_s/\rho(x_*)} \right) \\ &= \frac{G(x)}{G(y)} \frac{\sin \pi(x-y)}{\pi(x-y)} \end{aligned}$$

where $G(x) = e^{xx_*/2\rho(x_*)}$. This is what we wanted to show.

8. Edge fluctuations

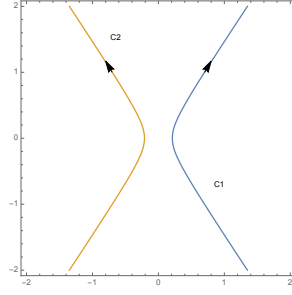
So what happens in case $x = \pm 2$? We will focus on $x = 2$.

Then

$$F'(1) = F''(1) = 0, \quad F'''(1) = 2.$$

This is a critical point of higher order. Now there are three paths of steepest descent and three paths of steepest ascent emanating from the saddle point (with angles $\pi/3$ between them)

We deform the contour γ_1 so that it passes through 1, incoming at at an angle $-\pi/3$ and leaving at at an angle $\pi/3$. We deform the contour γ_2 so that it passes through 1, incoming at at an angle $-2\pi/3$ and leaving at at an angle $2\pi/3$. Then γ_1 , resp. γ_2 , consists of the steepest descent, resp. ascent, path for $\text{Re}F(z)$.



If we set

$$x = 2 + \frac{u}{n^{2/3}}$$

and

$$y = 2 + \frac{v}{n^{2/3}}$$

And then rescale

$$z = 1 + \frac{s}{n^{1/3}} \quad w = 1 + \frac{t}{n^{1/3}}$$

Then

$$nF(z; x) = -\frac{3n}{2} - n^{1/3}u + \frac{s^3}{3} - us + \mathcal{O}(n^{-2/3}),$$

and

$$F(w; y) = -\frac{3n}{2} - n^{1/3}v + \frac{t^3}{3} - vt + \mathcal{O}(n^{-1/3}).$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{e^{n^{1/3}(v-u)}}{n^{2/3}} K_n\left(2 + \frac{u}{n^{2/3}}, 2 + \frac{v}{n^{2/3}}\right) = \frac{1}{(2\pi i)^2} \int_{C_1} ds \int_{C_2} dt \frac{e^{\frac{s^3}{3} - us - \frac{t^3}{3} + vt}}{s - t}$$

where the contours C_1 and C_2 are as in the picture on the right.

It remains to see that the right-hand side is a representation of the Airy kernel

$$K_{Airy}(u, v) = \frac{1}{(2\pi i)^2} \int_{C_1} ds \int_{C_2} dt \frac{e^{\frac{s^3}{3} - us - \frac{t^3}{3} + vt}}{s - t}$$

APPENDIX A

Laurent, Toeplitz, and Hankel operators

0.1. Laurent operators and symbols. Let $\ell^2(\mathbb{Z})$ be the Hilbert space of square-summable sequences $f = (f_k)_{k \in \mathbb{Z}}$. A (bi-infinite) matrix $L = (L_{ij})_{i,j \in \mathbb{Z}}$ is called a *Laurent matrix* if L_{ij} depends only on $i - j$, i.e.

$$L_{ij} = a_{i-j} \quad (i, j \in \mathbb{Z})$$

for some sequence $(a_k)_{k \in \mathbb{Z}}$. The corresponding bounded operator on $\ell^2(\mathbb{Z})$ (when it exists) is called a *Laurent operator* and is denoted $L(a)$, where a is the *symbol*

$$a(z) = \sum_{k \in \mathbb{Z}} a_k z^k, \quad z \in \mathbb{T}.$$

When $(a_k) \in \ell^1(\mathbb{Z})$, the series converges absolutely on $\mathbb{T}$ and $L(a)$ is bounded, acting by convolution:

$$(L(a)f)_i = \sum_{k \in \mathbb{Z}} a_{i-k} f_k.$$

In this case one has the algebra property

$$L(a)L(b) = L(ab),$$

reflecting the fact that Laurent operators diagonalize under the Fourier transform.

A convenient Banach algebra for symbols is the *Wiener algebra*

$$\mathcal{W} := \left\{ a(z) = \sum_{k \in \mathbb{Z}} a_k z^k : \sum_{k \in \mathbb{Z}} |a_k| < \infty \right\}, \quad \|a\|_{\mathcal{W}} = \sum_{k \in \mathbb{Z}} |a_k|.$$

If $a \in \mathcal{W}$ and $a(z) \neq 0$ for all $z \in \mathbb{T}$, then $1/a \in \mathcal{W}$ (Wiener's lemma). Consequently,

$$L(a)^{-1} = L(1/a)$$

on $\ell^2(\mathbb{Z})$. This is one reason why operator-level inverses are often simpler than finite-matrix inverses.

0.2. Toeplitz and Hankel operators. Let $\ell^2(\mathbb{Z}_{\geq 0})$ be the Hardy-type half-space, and let $P_+ : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ be the orthogonal projection onto indices ≥ 0 . For a Laurent operator $L(a)$, the associated *Toeplitz operator* is the compression

$$T(a) := P_+ L(a) P_+ \Big|_{\ell^2(\mathbb{Z}_{\geq 0})}.$$

In matrix form, $T(a)$ has entries

$$T(a)_{ij} = a_{i-j} \quad (i, j \geq 0),$$

so it is constant along diagonals. The *finite Toeplitz matrix* of size n is the further compression

$$T_n(a) := P_n T(a) P_n,$$

where P_n projects onto $\{0, 1, \dots, n-1\}$.

The *Hankel operator* associated with a is defined by

$$H(a) := P_+ L(a) J P_+ \Big|_{\ell^2(\mathbb{Z}_{\geq 0})},$$

where J is the flip $(Jf)_k = f_{-k-1}$ on $\ell^2(\mathbb{Z})$. In matrix form,

$$H(a)_{ij} = a_{i+j+1} \quad (i, j \geq 0),$$

so it is constant along anti-diagonals.

A fundamental algebraic identity relating these objects is

$$(23) \quad T(a)T(b) = T(ab) + H(a)H(\tilde{b}), \quad \tilde{b}(z) := b(z^{-1}),$$

valid under mild summability assumptions (e.g. $a, b \in \mathcal{W}$). The additional Hankel term measures the failure of the Toeplitz compression to be multiplicative.

0.3. Wiener–Hopf factorization and invertibility of Toeplitz operators. Write the Fourier decomposition

$$a(z) = a_-(z) a_+(z),$$

where

$$a_+(z) = \sum_{k \geq 0} a_k^{(+)} z^k, \quad a_-(z) = \sum_{k \leq 0} a_k^{(-)} z^k,$$

and both a_+ and a_- are nonvanishing on $\mathbb{T}$. Such a factorization is called a *Wiener–Hopf factorization*. In the Wiener algebra $\mathcal{W}$ one can formulate it as

$$a \in \mathcal{W}, \ a(z) \neq 0 \ \forall z \in \mathbb{T}, \ \text{wind}(a, 0) = 0 \implies a = a_- a_+, \quad a_{\pm}^{\pm 1} \in \mathcal{W},$$

where $\text{wind}(a, 0)$ is the winding number of $a(\mathbb{T})$ around 0. When $\text{wind}(a, 0) \neq 0$, a canonical factorization includes an extra monomial z^{κ} .

When $a = a_- a_+$ with $a_{\pm}^{\pm 1} \in \mathcal{W}$ and $\text{wind}(a, 0) = 0$, the Toeplitz operator $T(a)$ is invertible on $\ell^2(\mathbb{Z}_{\geq 0})$, and one has the operator identity

$$(24) \quad T(a)^{-1} = T(a_+^{-1}) T(a_-^{-1}) \quad \text{modulo a compact correction (in fact Hankel-type)}.$$

More precisely, using (23) one finds that $T(a_+^{-1})T(a_-^{-1})$ is a two-sided inverse of $T(a)$ up to a correction expressed through Hankel operators built from $a_{\pm}^{\pm 1}$. This is the analytic core of the Wiener–Hopf method for Toeplitz operators.

0.4. Finite Toeplitz matrices and approximate inverses as $n \rightarrow \infty$.

Even when $T(a)$ is easy to invert at the operator level, the explicit inverse of the finite matrix $T_n(a)$ can be difficult. A useful viewpoint is that $T_n(a)$ is a finite section of $T(a)$, and $T_n(a)^{-1}$ should be close to the corresponding finite section of $T(a)^{-1}$, up to boundary effects near 0 and $n-1$.

Assume $a \in \mathcal{W}$ is nonvanishing on $\mathbb{T}$ with $\text{wind}(a, 0) = 0$, and choose a Wiener–Hopf factorization $a = a_- a_+$ with $a_{\pm}^{\pm 1} \in \mathcal{W}$. Then the following principles are standard.

(1) Finite section method. Under the assumptions above, the finite sections are stable:

$$T_n(a) \text{ is invertible for all sufficiently large } n, \quad \sup_{n \geq n_0} \|T_n(a)^{-1}\| < \infty.$$

Moreover, in a strong sense,

$$T_n(a)^{-1} \approx P_n T(a)^{-1} P_n \quad (n \rightarrow \infty),$$

with an error concentrated near the boundary (hence effectively low-rank after suitable truncations).

(2) Gohberg–Semencul-type structure. There are explicit representations of $T_n(a)^{-1}$ in terms of Toeplitz and Hankel matrices built from the Fourier coefficients of $a_{\pm}^{\pm 1}$, showing that

$$T_n(a)^{-1} = T_n(a_+^{-1}) T_n(a_-^{-1}) + (\text{boundary correction}),$$

where the boundary correction has small rank compared to n (often rank at most $2r$ for symbols with finite bandwidth, and more generally a correction whose operator norm decays as $n \rightarrow \infty$ under additional regularity/analyticity).

(3) Exponentially small errors for analytic symbols. If a extends analytically and nonvanishingly to an annulus $\{r < |z| < R\}$ with $r < 1 < R$, then the Fourier coefficients of $a_{\pm}^{\pm 1}$ decay exponentially, and the boundary correction above becomes exponentially small in n (in operator norm or entrywise, depending on the formulation). This is the regime in which approximate inverses are particularly effective.

In applications, one typically combines the operator factorization (which yields a manageable expression for $T(a)^{-1}$) with one of the finite- n approximation principles above. This allows one to control kernels or partition functions in the limit $n \rightarrow \infty$ without requiring an explicit closed form for $T_n(a)^{-1}$.

0.5. The special case of rational symbols. An important and particularly tractable situation arises when the symbol $a(z)$ is a rational function of z . In this case many of the abstract constructions discussed above admit explicit realizations, and several difficulties present for general symbols disappear.

Assume that $a(z)$ is rational and has no zeros or poles on the unit circle. Then a automatically belongs to the Wiener algebra $\mathcal{W}$, and its Wiener–Hopf factorization can be carried out explicitly by separating the zeros and poles inside and outside the unit disk. More precisely, one can write

$$a(z) = c z^{\kappa} \prod_{j=1}^J \frac{1 - \overline{\alpha_j} z}{1 - \alpha_j z} \prod_{k=1}^K \frac{1 - \beta_k z^{-1}}{1 - \overline{\beta_k} z^{-1}},$$

where $|\alpha_j| < 1$, $|\beta_k| < 1$, $c \neq 0$, and $\kappa \in \mathbb{Z}$ is the winding number of a . From this representation the Wiener–Hopf factors a_+ and a_- are obtained by grouping together the factors that are analytic inside and outside the unit disk, respectively.

In the rational case the associated Toeplitz and Hankel operators have additional structure. In particular, the Hankel operators $H(a)$ and $H(\tilde{a})$ have finite rank, with rank equal to the total number of poles of a inside or outside the unit disk. As a consequence, the correction terms appearing in identities such as

$$T(a)^{-1} = T(a_+^{-1}) T(a_-^{-1}) + (\text{Hankel correction})$$

are finite-rank operators.

This finite-rank property has two important implications. First, the inverse Toeplitz operator $T(a)^{-1}$ differs from the simple product $T(a_+^{-1})T(a_-^{-1})$ only by an explicitly computable, finite-dimensional correction. Second, for finite Toeplitz matrices $T_n(a)$, the boundary corrections in the approximate inverse are uniformly finite rank, independent of n . In particular, away from the boundary the entries of $T_n(a)^{-1}$ stabilize exactly once n is sufficiently large.

In asymptotic problems, and especially in determinantal point process limits, rational symbols therefore lead to kernels that are finite-rank perturbations of translation-invariant kernels. This feature is often crucial for explicit computations and for identifying integrable structures in the limiting process.

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